

Diagnosis Questions

Category: Diagnosis **Total Questions:** 206 **Purpose:** Medical professional review of categorization accuracy

Review Instructions

For each question below, please mark:

- ✓ **Correctly categorized** - Question clearly belongs in this category
- ✗ **Miscategorized** - Question belongs in a different category (note which one)
- ? **Ambiguous** - Question could reasonably fit multiple categories

You don't need to review all questions! Even 20-30 per category is very helpful.

What 'Diagnosis' Means

Questions asking you to identify the disease, condition, or underlying cause of the patient's presentation.

Questions for Review

Question 1

Clinical Scenario:

A 73-year-old man noted a rapid onset of severe dizziness and difficulty swallowing while watching TV at home. His wife reports that he had difficulty forming sentences and his gait was unsteady at this time. Symptoms were severe within 1 minute and began to improve spontaneously after 10 minutes. He has had type 2 diabetes mellitus for 25 years and has a 50 pack-year smoking history. On arrival to the emergency department 35 minutes after the initial development of symptoms, his manifestations have largely resolved with the exception of a subtle nystagmus and ataxia. His blood pressure is 132/86 mm Hg, the heart rate is 84/min, and the respiratory rate is 15/min. After 45 minutes, his symptoms are completely resolved, and neurological examination is unremarkable. Which of the following is the most likely cause of this patient's condition?

Answer Choices:

A. Vertebral artery occlusion B. Middle cerebral artery occlusion C. Posterior cerebral artery occlusion D. Lenticulostriate artery occlusion

Correct Answer: A. Vertebral artery occlusion

Question 2

Clinical Scenario:

A 60-year-old woman and her son are visited at her home by a health aid. He is her caregiver but has difficulty getting her out of the house. Her son is concerned about continuous and repetitive mouth and tongue movements that started about 2 weeks ago and have become more evident ever since. She is non-verbal at baseline and can complete most activities of daily living. She suffers from an unspecified psychiatric disorder. Her medications include fluphenazine. Today, her heart rate is 90/min, respiratory rate is 17/min, blood pressure is 125/87 mm Hg, and temperature is 37.0°C (98.6°F). On physical exam, she appears gaunt and anxious. She is drooling and her mouth is making a chewing motion that is occasionally disrupted by wagging her tongue back and forth. She seems to be performing these motions absentmindedly. Her heart has a regular rate and rhythm and her lungs are clear to auscultation bilaterally. CMP, CBC, and TSH are normal. A urine toxicology test is negative. What is the next best step in her management?

Answer Choices:

A. Start clozapine B. Switch to chlorpromazine C. Expectant management D. Stop the medication

Correct Answer: D. Stop the medication

Question 3

Clinical Scenario:

A 35-year-old female presents to her PCP at the request of her husband after 3 weeks of erratic behavior. The patient has been staying up all night online shopping on eBay. Despite a lack of sleep, she is "full of energy" during the day at her teaching job, which she believes is "beneath [her], anyway." She has not sought psychiatric treatment in the past, but reports an episode of self-diagnosed depression 2 years ago. The patient denies thoughts of suicide. Pregnancy test is negative. Which of the following is the best initial treatment?

Answer Choices:

A. Valproate and venlafaxine B. Valproate and olanzapine C. Haloperidol D. Electroconvulsive therapy

Correct Answer: B. Valproate and olanzapine

Question 4

Clinical Scenario:

A 63-year-old man undergoes workup for nocturnal dyspnea and what he describes as a "choking" sensation while sleeping. He also endorses fatigue and dyspnea on exertion. Physical exam reveals a normal S1, loud P2, and a neck circumference of 17 inches (43 cm) (normal < 14 inches (< 35 cm)). His temperature is 98.8°F (37°C), blood pressure is 128/82 mmHg, pulse is 86/min, and respirations are 19/min. He undergoes spirometry, which is unrevealing, and polysomnography, which shows 16 hypopneic and apneic events per

hour. Mean pulmonary arterial pressure is 30 mmHg. Which of the following complications is this patient most at risk for?

Answer Choices:

A. Chronic obstructive pulmonary disease B. Left ventricular failure C. Pulmonary embolism D. Right ventricular failure

Correct Answer: D. Right ventricular failure

Question 5

Clinical Scenario:

A 6-month-old girl presents to an outpatient office for a routine physical. She can sit momentarily propped on her hands, babbles with consonants, and transfers objects hand to hand. The pediatrician assures the parents that their daughter has reached appropriate developmental milestones. Which additional milestone would be expected at this stage in development?

Answer Choices:

A. Engaging in pretend play B. Separation anxiety C. Showing an object to her parents to share her interest in that object D. Stranger anxiety

Correct Answer: D. Stranger anxiety

Question 6

Clinical Scenario:

A 24-year-old previously healthy man comes to his physician because of dyspnea and hemoptysis for the past week. Examination shows inspiratory crackles at both lung bases. The remainder of the examination shows no abnormalities. His hemoglobin concentration is 14.2 g/dL, leukocyte count is 10,300/mm³, and platelet count is 205,000/mm³. Urine dipstick shows 2+ proteins. Urinalysis shows 80 RBC/hpf and 1–2 WBC/hpf. An x-ray of the chest shows pulmonary infiltrates. Further evaluation is most likely to show increased serum titers of which of the following?

Answer Choices:

A. Anti-GBM antibody B. Anti-dsDNA antibody C. Anti-PLA2R antibody D. P-ANCA

Correct Answer: A. Anti-GBM antibody

Question 7

Clinical Scenario:

A 60-year-old man who is a chronic smoker comes to the hospital with the chief complaint of shortness of breath which has worsened over the past 2 days. He also has a productive cough with yellowish sputum. There is no history of hemoptysis, chest pain, fever, palpitations, or leg swelling. He had a viral illness one week ago. He has been using an inhaler for 10 years for his respiratory disease. He sleeps with 2 pillows every night. He received 100 mg of hydrocortisone and antibiotics in the emergency department, and his symptoms have subsided. His FEV1/FVC ratio is < 0.70 , and FEV1 is 40% of predicted. What is the most likely finding that can be discovered from the histology of his bronchioles?

Answer Choices:

A. Curschmann spirals B. Increase Reid index C. Ferruginous bodies D. Non-caseating granuloma

Correct Answer: B. Increase Reid index

Question 8

Clinical Scenario:

A 71-year-old man presents to his oncologist with nausea. He recently underwent chemotherapy for pancreatic cancer and has developed severe intractable nausea over the past week. He vomits several times a day. His past medical history is notable for gout, osteoarthritis, and major depressive disorder. He takes allopurinol and sertraline. He has a 15-pack-year smoking history and drinks 1 glass of wine per day. His temperature is 98.6°F (37°C), blood pressure is 148/88 mmHg, pulse is 106/min, and respirations are 22/min. On exam, he is lethargic but able to answer questions appropriately. He has decreased skin turgor and dry mucous membranes. He is started on a medication to treat nausea. However, 3 days later he presents to the emergency room with fever, agitation, hypertonia, and clonus. What is the most likely mechanism of action of the drug this patient was prescribed?

Answer Choices:

A. 5-HT₃ receptor antagonist B. D₂ receptor antagonist C. H₁ receptor antagonist D. M₁ receptor antagonist

Correct Answer: A. 5-HT₃ receptor antagonist

Question 9

Clinical Scenario:

A 26-year-old G1P0 female who is 39 weeks pregnant presents to the emergency department in labor. She reports following her primary care physician's recommendations throughout her pregnancy and has not had any complications. During delivery, the baby's head turtled back into the vaginal canal and did not advance any further. The neonatal intensivist was called for shoulder dystocia and a baby girl was able to be delivered vaginally 6 minutes later. Upon initial assessment, the baby appeared pale throughout, had her arms and legs flexed without active motion, and had some flexion of extremities when stimulated. Her pulse is 120/min and had irregular respirations. What is this baby's initial APGAR score?

Answer Choices:

A. 3 B. 4 C. 5 D. 6

Correct Answer: C. 5

Question 10

Clinical Scenario:

A 62-year-old woman with hypertension and type 2 diabetes mellitus comes to the physician because of increasing shortness of breath and a dry cough over the past 6 months. She has smoked 1 pack of cigarettes daily for the past 40 years. Chest auscultation shows scattered expiratory wheezes in both lung fields. Spirometry shows an FEV1:FVC ratio of 65% and an FEV1 of 70% of predicted. Her diffusing capacity for carbon monoxide (DLCO) is 42% of predicted. Which of the following is the most likely diagnosis?

Answer Choices:

A. Pulmonary fibrosis B. Emphysema C. Bronchial asthma D. Bronchiectasis

Correct Answer: B. Emphysema

Question 11

Clinical Scenario:

A 69-year-old woman is brought to the physician by her husband because of multiple falls and difficulty maintaining balance while standing or walking over the past year. During this period, she has had blurred vision and diplopia. Her husband has had difficulty understanding her speech for the past 3 months. She has become withdrawn and now refuses to go to social gatherings. Examination shows a broad-based gait and dysarthria. The visual acuity is 20/20 in each eye. There is conjugate limitation of both eyes while looking down. Muscle tone is increased in bilateral upper extremities. Bradykinesia is present. Mental status examination shows apathy. She responds to questions with 1–2 words after a delay of several seconds. Grasp reflex is present. An MRI of the brain is most likely to show which of the following?

Answer Choices:

A. Asymmetric focal cortical atrophy B. Midbrain atrophy with intact pons C. Frontal atrophy with intact hippocampi D. Enlarged ventricles with mild cortical atrophy

Correct Answer: B. Midbrain atrophy with intact pons

Question 12

Clinical Scenario:

A 2-year-old boy is brought to the physician by his parents because of difficulty walking and cold feet for the past 2 months. His parents report that he tires quickly from walking. The patient was born at 37 weeks' gestation and has met all developmental milestones. There is no personal or family history of serious illness.

He is at the 50th percentile for height and 40th percentile for weight. His temperature is 36.9°C (98.4°F), pulse is 119/min, respirations are 32/min, and blood pressure is 135/85 mm Hg. A grade 2/6 systolic murmur is heard in the left paravertebral region. Pedal pulses are absent. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Rib notching B. Low tissue oxygenation in the legs C. Interarm difference in blood pressure D. Right ventricular outflow obstruction

Correct Answer: B. Low tissue oxygenation in the legs

Question 13

Clinical Scenario:

A 34-year-old man comes to the physician because of increasing lower back and neck pain for the past 7 months. The pain is worse in the morning and improves when he plays basketball. He has noticed shortness of breath while playing for the past 2 months. He is sexually active with 2 female partners and uses condoms inconsistently. He appears lethargic. His vital signs are within normal limits. Examination of the back shows tenderness over the sacroiliac joints. Range of motion is limited. The lungs are clear to auscultation. Chest expansion is decreased on full inspiration. His leukocyte count is 14,000/mm³ and erythrocyte sedimentation rate is 84 mm/h. An x-ray of the spine shows erosion and sclerosis of the sacroiliac joints and loss of spinal lordosis. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Paresthesia over the anterolateral part of the thigh B. Nail pitting and separation of the nail from the nailbed C. Tenderness at the Achilles tendon insertion site D. Ulnar deviation of the fingers bilaterally

Correct Answer: C. Tenderness at the Achilles tendon insertion site

Question 14

Clinical Scenario:

A 69-year-old man presents to the emergency department with shortness of breath that has been worsening over the past month. The patient states that he has had worsening shortness of breath that has progressed to shortness of breath with simple activities and minor exertion. When he was unable to climb the stairs to his house today, he decided to come to the emergency department. The patient has a past medical history of poorly managed diabetes mellitus, hypertension, end stage renal disease, and obesity. His current medications include insulin, metformin, lisinopril, hydrochlorothiazide, and ibuprofen. The patient is notably non-compliant with his medications. An EKG and chest radiograph are normal. The patient had dialysis two days ago and attends most of his appointments. Laboratory values are ordered and are seen below:

Serum: Na⁺: 135 mEq/L K⁺: 4.5 mEq/L Cl⁻: 100 mEq/L HCO₃⁻: 19 mEq/L Urea nitrogen: 29 mg/dL Glucose: 75 mg/dL Creatinine: 2.5 mg/dL Ca²⁺: 9.2 mg/dL Mg²⁺: 1.7 mEq/L AST: 11 U/L ALT: 11 U/L

Leukocyte count and differential: Leukocyte count: 4,500/mm³ Platelet count: 150,000/mm³ Neutrophil: 54% Lymphocyte: 25% Monocyte: 3% Eosinophil: 1% Basophil: 1%

Hemoglobin: 8.2 g/dL Hematocrit: 22% Mean corpuscular volume: 82 μm^3

The patient appears comfortable at rest but demonstrates notable shortness of breath when exerting himself. His temperature is 99.5°F (37.5°C), pulse is 89/min, blood pressure is 144/85 mmHg, respirations are 10/min, and oxygen saturation is 97% on room air. Pulmonary and cardiac exam are within normal limits. Which of the following is a side-effect of the long-term therapy this patient should be started on right now?

Answer Choices:

A. Hypertension B. Hyperkalemia C. Hypokalemia D. Visual halos

Correct Answer: A. Hypertension

Question 15

Clinical Scenario:

A 36-year-old woman is brought to the emergency department 20 minutes after being involved in a high-speed motor vehicle collision. On arrival, she is unconscious. Her pulse is 140/min, respirations are 12/min and shallow, and blood pressure is 76/55 mm Hg. 0.9% saline infusion is begun. A focused assessment with sonography shows blood in the left upper quadrant of the abdomen. Her hemoglobin concentration is 7.6 g/dL and hematocrit is 22%. The surgeon decided to move the patient to the operating room for an emergent explorative laparotomy. Packed red blood cell transfusion is ordered prior to surgery. However, a friend of the patient asks for the transfusion to be held as the patient is a Jehovah's Witness. The patient has no advance directive and there is no documentation showing her refusal of blood transfusions. The patient's husband and children cannot be contacted. Which of the following is the most appropriate next best step in management?

Answer Choices:

A. Administer hydroxyethyl starch B. Transfusion of packed red blood cells C. Consult hospital ethics committee D. Proceed to surgery without transfusion

Correct Answer: B. Transfusion of packed red blood cells

Question 16

Clinical Scenario:

A 25-year-old homeless woman presents to an urgent care clinic complaining of vaginal bleeding. She also has vague lower right abdominal pain which started a few hours ago and is increasing in intensity. The medical history is significant for chronic hepatitis C infection, and she claims to take a pill for it 'every now and then.' The temperature is 36.0°C (98.6°F), the blood pressure is 110/70 mmHg, and the pulse is 80/min. The abdominal examination is positive for localized right adnexal tenderness; no rebound tenderness or guarding is noted. A transvaginal ultrasound confirms a 2.0 cm gestational sac in the right fallopian tube. What is the next appropriate step in the management of this patient? Immunodeficiency (RA, SLE, and Crohns)

Answer Choices:

A. Surgery B. IV fluids, then surgery C. Methotrexate D. Tubal ligation

Correct Answer: A. Surgery

Question 17**Clinical Scenario:**

A 4-year-old boy presents with a dry cough. The patient's mother states that the cough started a week ago and has not improved. She says the patient will have fits of forceful coughing that will last for minutes, followed by gasping as he catches his breath. Occasionally, the patient will vomit after one of these episodes. Past medical history is significant for a recent upper respiratory infection 4 weeks ago that has resolved. No current medications. Patient immunization status is incomplete because his mother believes they are harmful. Vitals are temperature 37.0°C (98.6°F), blood pressure 105/65 mm Hg, pulse 101/min, respiratory rate 27/min, and oxygen saturation 99% on room air. Cardiac exam is normal. Lungs are clear to auscultation. There are conjunctival hemorrhages present bilaterally. Which of the following correctly describes the stage of this patient's most likely diagnosis?

Answer Choices:

A. Catarrhal stage B. Paroxysmal stage C. Convalescent stage D. Intermittent stage

Correct Answer: B. Paroxysmal stage

Question 18**Clinical Scenario:**

A 29-year-old G1P0 female at 32 weeks gestation presents to the emergency department with vaginal bleeding. She has had minimal prenatal care to-date with only an initial visit with an obstetrician after a positive home pregnancy test. She describes minimal spotting that she noticed earlier today that has progressed to larger amounts of blood; she estimates 30 mL of blood loss. She denies any cramping, pain, or contractions, and she reports feeling continued movements of the baby. Ultrasound and fetal heart rate monitoring confirm the presence of a healthy fetus without any evidence of current or impending complications. The consulted obstetrician orders blood testing for Rh-status of both the mother as well as the father, who brought the patient to the hospital. Which of the following represents the best management strategy for this situation?

Answer Choices:

A. If mother is Rh-negative and father is Rh-positive then administer RhoGAM B. If mother is Rh-positive and father is Rh-negative then administer RhoGAM C. If mother is Rh-negative and father is Rh-positive, RhoGAM administration is not needed D. After 28 weeks gestation, administration of RhoGAM will have no benefit

Correct Answer: A. If mother is Rh-negative and father is Rh-positive then administer RhoGAM

Question 19

Clinical Scenario:

A 63-year-old man presents to his primary care provider complaining of changes in his vision. He says that he has been having transient episodes of "shimmering lights" and generalized blurring of his vision for the past 3 months. He is disturbed by this development as he worries it may interfere with his job as a bus driver. He additionally reports a 12-pound weight loss over this time unaccompanied by a change in appetite, and his gout flares have grown more frequent despite conforming to his recommended diet and allopurinol. His temperature is 98.0°F (36.7°C), blood pressure is 137/76 mmHg, pulse is 80/min, and respirations are 18/min. Hemoglobin and hematocrit obtained the previous day were 18.1 g/dL and 61%, respectively. Peripheral blood screening for JAK2 V617F mutation is positive. Which of the following findings is most likely expected in this patient?

Answer Choices:

A. Decreased erythrocyte sedimentation rate B. Increased erythropoietin levels C. Schistocytes on peripheral smear D. Thrombocytopenia

Correct Answer: A. Decreased erythrocyte sedimentation rate

Question 20

Clinical Scenario:

A 74-year-old man is brought to the emergency department because of lower abdominal pain for 3 hours. The pain is sharp, constant, and does not radiate. He has not urinated for 24 hours and he has not passed stool for over 3 days. He was diagnosed with herpes zoster 3 weeks ago and has been taking amitriptyline for post-herpetic neuralgia for 1 week. Last year he was diagnosed with nephrolithiasis and was treated with lithotripsy. He has a history of hypertension, benign prostatic hyperplasia, and coronary artery disease. His other medications include amlodipine, metoprolol, tamsulosin, aspirin, and simvastatin. He appears uncomfortable. His temperature is 37.3°C (99.1°F), pulse is 102/min, and blood pressure is 140/90 mm Hg. Abdominal examination shows a palpable lower midline abdominal mass that is tender to palpation. Bowel sounds are hypoactive. The remainder of the examination shows no abnormalities. A pelvic ultrasound shows an anechoic mass in the lower abdomen. Which of the following is the most appropriate next step in the management of this patient?

Answer Choices:

A. Observation and NSAIDs administration B. Transurethral catheterization C. Finasteride administration D. IV pyelography

Correct Answer: B. Transurethral catheterization

Question 21

Clinical Scenario:

A 40-year-old woman comes to the physician for a 2-month history of chest pain and heartburn after meals. The patient reports that the pain is worse at night and especially when lying down. She has a history of Raynaud's disease treated with nifedipine. There is no family history of serious illness. She emigrated to the US from Nigeria 5 years ago. She does not smoke or drink alcohol. Vital signs are within normal limits. Cardiopulmonary examination shows no abnormalities. Thickening and hardening of the skin is seen on the hands and face. There are several firm, white nodules on the elbows and fingertips. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Anti-RNA polymerase III antibodies B. Anti-U1 RNP antibodies C. Anticentromere antibodies D. Anti-Scl-70 antibodies

Correct Answer: C. Anticentromere antibodies

Question 22

Clinical Scenario:

A 50-year-old man arrives to the clinic complaining of progressive weakness. He explains that for 3 months he has had difficulty climbing the stairs, which has now progressed to difficulty getting out of a chair. He denies diplopia, dysphagia, dyspnea, muscle aches, or joint pains. He denies weight loss, weight gain, change in appetite, or heat or cold intolerance. He reports intermittent low-grade fevers. He has a medical history significant for hypertension and hyperlipidemia. He has taken simvastatin and losartan daily for the past 6 years. His temperature is 99.0°F (37.2°C), blood pressure is 135/82 mmHg, and pulse is 76/min. Cardiopulmonary examination is normal. The abdomen is soft, non-tender, non-distended, and without hepatosplenomegaly. Muscle strength is 3/5 in the hip flexors and 4/5 in the deltoids, biceps, triceps, patellar, and Achilles tendon reflexes are 2+ and symmetric. Sensation to pain, light touch, and vibration are intact. Gait is cautious, but grossly normal. There is mild muscle tenderness of his thighs and upper extremities. There is no joint swelling or erythema and no skin rashes. A complete metabolic panel is within normal limits. Additional lab work is obtained as shown below:

Serum: Na⁺: 141 mEq/L Cl⁻: 100 mEq/L K⁺: 4.3 mEq/L HCO₃⁻: 23 mEq/L Urea nitrogen: 18 mg/dL Glucose: 128 mg/dL Creatinine: 1.0 mg/dL Alkaline phosphatase: 69 U/L Aspartate aminotransferase (AST): 302 U/L Alanine aminotransferase (ALT): 210 U/L TSH: 6.9 uU/mL Thyroxine (T4): 5.8 ug/dL Creatine kinase: 4300 U/L C-reactive protein: 11.9 mg/L Erythrocyte sedimentation rate: 37 mm/h

Which of the following is the most accurate diagnostic test?

Answer Choices:

A. Autoantibodies B. Electromyography C. Muscle biopsy D. Statin cessation

Correct Answer: C. Muscle biopsy

Question 23

Clinical Scenario:

A 12-year-old boy is brought to the emergency department by his parents after he was bitten by a friend's cat while playing at their house. The patient reports moderate pain of the right hand and wrist but has full range of motion and strength. He is up to date on his vaccinations and is generally healthy. His vitals are unremarkable. Physical exam reveals a deep puncture wound that is actively bleeding. The wound is irrigated and a dressing is applied. Which of the following is appropriate management of this patient?

Answer Choices:

A. Ampicillin-sulbactam, surgical debridement, and laceration closure B. Amoxicillin-clavulanate C. Amoxicillin-clavulanate and laceration closure D. Laceration closure

Correct Answer: B. Amoxicillin-clavulanate

Question 24

Clinical Scenario:

A 34-year-old female presents to her primary care physician complaining of fatigue. Over the last three months she has experienced decreased energy and gained 7 pounds. Review of systems is negative for symptoms of depression but is positive for constipation, myalgias, and cold intolerance. Physical exam is notable for delayed deep tendon reflex relaxation. Vital signs are as follows: T 37.1 C, HR 61, BP 132/88, RR 16, and SpO2 100%. Which of the following is the best initial screening test for this patient?

Answer Choices:

A. Level of anti-thyroid peroxidase (TPO) antibodies B. Morning cortisol and plasma ACTH C. Serum TSH D. Hemoglobin and hematocrit

Correct Answer: C. Serum TSH

Question 25

Clinical Scenario:

A 28-year-old man comes to the physician because of progressively worsening fatigue, nausea, and right upper quadrant pain. He has a history of intravenous heroin use. Both serum Anti-HBe and Anti-HBc are positive. Further analysis of the Anti-HBc immunoglobulin is most likely to show which of the following properties?

Answer Choices:

A. Activates eosinophils B. Forms a pentamer when secreted C. Protects against gastrointestinal infections D. Crosses the placenta

Correct Answer: B. Forms a pentamer when secreted

Question 26

Clinical Scenario:

A 60-year-old woman presents to the dermatologist with a lesion on her lower eyelid. She noticed it a month ago and looked like a pimple. She says that it has been bleeding lately with minimal trauma which alarmed her. She says the lesion has not grown in size and is not associated with pain or pruritus. No significant past medical history. Physical examination reveals a 0.5 cm lesion that has a pearly appearance with telangiectasia and central ulceration and curled borders. The lesion is biopsied. Histopathology reveals peripheral palisading cells with large, hyperchromatic nuclei and a high nuclear: cytoplasmic ratio. Which of the following mechanisms best describes the most common mode of spread of this patient's neoplasm?

Answer Choices:

A. Seeding B. Does not spread (tumor is typically benign) C. Local invasion via collagenase D. Lymphatic spread

Correct Answer: C. Local invasion via collagenase

Question 27

Clinical Scenario:

A 32-year-old woman comes to the physician for a screening health examination that is required for scuba diving certification. The physician asks her to perform a breathing technique: following deep inspiration, she is instructed to forcefully exhale against a closed airway and contract her abdominal muscles while different cardiovascular parameters are evaluated. Which of the following effects is most likely after 10 seconds in this position?

Answer Choices:

A. Decreased systemic vascular resistance B. Decreased left ventricular stroke volume C. Decreased intra-abdominal pressure D. Decreased pulse rate

Correct Answer: B. Decreased left ventricular stroke volume

Question 28

Clinical Scenario:

A 7-year-old boy is brought to the pediatrician by his parents for a routine checkup. The parents note that the patient recently joined a baseball team and has had trouble keeping up with his teammates and gets short of breath with exertion. The patient has otherwise been healthy and has no known history of asthma or allergic reaction. Today, the patient's temperature is 98.2°F (36.8°C), blood pressure is 112/72 mmHg, pulse is 70/min, and respirations are 12/min. The physical exam is notable for a heart murmur that decreases when the patient bears down. Additionally, the hand grip and rapid squatting maneuvers increase the severity of the murmur. Which of the following is likely heard on auscultation?

Answer Choices:

A. Continuous murmur inferior to the left clavicle B. Crescendo-decrescendo systolic murmur radiating to carotids C. Holosystolic murmur at the apex radiating to the axilla D. Holosystolic murmur at the lower left sternal border

Correct Answer: D. Holosystolic murmur at the lower left sternal border

Question 29**Clinical Scenario:**

A 32-year-old woman with a recurrent vesicular genital rash comes to the physician because of a 3-day history of a painful, pruritic rash that began on the extremities and has spread to her trunk. Her only medication is acyclovir. Her temperature is 38.1°C (100.6°F). Examination of the skin shows several reddish-purple papules and macules, some of which have a dusky center with a lighter ring around them. Which of the following is the most likely diagnosis?

Answer Choices:

A. Erythema multiforme B. Urticaria C. Stevens-Johnson syndrome D. Dermatitis herpetiformis

Correct Answer: A. Erythema multiforme

Question 30**Clinical Scenario:**

A 44-year-old man presents to a clinic for the evaluation of difficulty swallowing for the past few days. He says that he has noticed progressively worsening chest pain when he attempts to swallow solids or liquids. He works from a home office, has not had any recent sick contacts, and is currently not sexually active. His medical history includes AIDS. His current medications include emtricitabine, rilpivirine, and tenofovir. His temperature is 38.1°C (100.6°F), pulse is 72/min, respirations are 18/min, and blood pressure is 136/84 mm Hg. A physical examination is notable for a dry mouth with red mucosa and no distinct plaques or patches, and a supple neck with no masses or cervical lymphadenopathy. An esophagogastroduodenoscopy shows small white patches within the esophageal lumen. A biopsy of one of the lesions is performed and the microscopic appearance of the finding is shown below. Which of the following is the most likely diagnosis?

Answer Choices:

A. Cytomegalovirus esophagitis B. Esophageal candidiasis C. Herpes esophagitis D. Medication-induced esophagitis

Correct Answer: B. Esophageal candidiasis

Question 31

Clinical Scenario:

A 45-year-old woman is brought to the Emergency Department by her husband due to increasing confusion. He reports that she has been urinating a lot for the past month or so, especially at night, and has also been constantly drinking water and tea. Lately, she has been more tired than usual as well. Her past medical history is significant for bipolar disorder. She takes lithium and a multivitamin. She has a levonorgestrel IUD. Her blood pressure is 140/90 mmHg, pulse rate is 95/min, respiratory rate is 16/min, and temperature is 36°C (96.8°F). At physical examination, she is drowsy and disoriented. Her capillary refill is delayed and her mucous membranes appear dry. The rest of the exam is nondiagnostic. Laboratory studies show: Na⁺: 148 mEq/L K⁺: 4.2 mEq/L Serum calcium: 11.0 mg/dL Creatinine: 1.0 mg/dL Urine osmolality: 190 mOsm/kg Serum osmolality: 280 mOsm/kg Finger-stick glucose: 120 mg/dL Fluid resuscitation is initiated. Which of the following is the most likely diagnosis?

Answer Choices:

A. SIADH B. Psychogenic polydipsia C. Nephrogenic diabetes insipidus D. Central diabetes insipidus

Correct Answer: C. Nephrogenic diabetes insipidus

Question 32

Clinical Scenario:

A 68-year-old man is brought to the emergency department because of a severe headache, nausea, and vomiting for 30 minutes. Forty-five minutes ago, he fell and struck his head, resulting in loss of consciousness for 1 minute. After regaining consciousness, he felt well for 15 minutes before the headache began. On arrival, the patient becomes rigid and his eyes deviate to the right; he is incontinent of urine. Intravenous lorazepam is administered and the rigidity resolves. Which of the following is the most likely cause of the patient's condition?

Answer Choices:

A. Cerebrospinal fluid production/absorption mismatch B. Rupture of bridging veins C. Acute insufficiency of cerebral blood flow D. Bleeding between dura mater and skull

Correct Answer: D. Bleeding between dura mater and skull

Question 33

Clinical Scenario:

A 62-year-old female with a history of uncontrolled hypertension undergoes kidney transplantation. One month following surgery she has elevated serum blood urea nitrogen and creatinine and the patient complains of fever and arthralgia. Her medications include tacrolimus and prednisone. If the patient were experiencing acute, cell-

mediated rejection, which of the following would you most expect to see upon biopsy of the transplanted kidney?

Answer Choices:

A. Sloughing of proximal tubular epithelial cells B. Lymphocytic infiltrate of the tubules and interstitium C. Drug precipitation in the renal tubules D. Granular immunofluorescence around the glomerular basement membrane

Correct Answer: B. Lymphocytic infiltrate of the tubules and interstitium

Question 34

Clinical Scenario:

A 32-year-old homeless woman is brought to the emergency department by ambulance 30 minutes after the police found her on the sidewalk. On arrival, she is unresponsive. Her pulse is 76/min, respirations are 6/min, and blood pressure is 110/78 mm Hg. Examination shows cool, dry skin. The pupils are pinpoint and react sluggishly to light. Intravenous administration of a drug is initiated. Two minutes after treatment is started, the patient regains consciousness and her respirations increase to 12/min. The drug that was administered has the strongest effect on which of the following receptors?

Answer Choices:

A. Ryanodine receptor B. μ -receptor C. GABAA receptor D. 5-HT_{2A} receptor

Correct Answer: B. μ -receptor

Question 35

Clinical Scenario:

A 23-year-old male presents to his primary care physician after an injury during a rugby game. The patient states that he was tackled and ever since then has had pain in his knee. The patient has tried NSAIDs and ice to no avail. The patient has no past medical history and is currently taking a multivitamin, fish oil, and a whey protein supplement. On physical exam you note a knee that is heavily bruised. It is painful for the patient to bear weight on the knee, and passive motion of the knee elicits some pain. There is laxity at the knee to varus stress. The patient is wondering when he can return to athletics. Which of the following is the most likely diagnosis?

Answer Choices:

A. Medial collateral ligament tear B. Lateral collateral ligament tear C. Anterior cruciate ligament tear D. Posterior cruciate ligament tear

Correct Answer: B. Lateral collateral ligament tear

Question 36

Clinical Scenario:

A young man about to leave for his freshman year of college visits his physician in order to ensure that his immunizations are up-to-date. Because he is living in a college dormitory, his physician gives him a vaccine that prevents meningococcal disease. What type of vaccine did this patient likely receive?

Answer Choices:

A. Live, attenuated B. Killed, inactivated C. Toxoid D. Conjugated polysaccharide

Correct Answer: D. Conjugated polysaccharide

Question 37

Clinical Scenario:

An investigator performs a twin study to evaluate the effects of a novel drug that decreases serum glucose by inhibiting a transporter on the basolateral membrane of proximal convoluted tubule cells. The results of the study are shown. Test Control Serum glucose (mg/dL) 82.4 99 Dipstick urine glucose negative negative Urine anion gap positive negative The drug most likely inhibits transport of which of the following substrates?"

Answer Choices:

A. Glutamine B. Sodium C. Alanine D. Leucine

Correct Answer: A. Glutamine

Question 38

Clinical Scenario:

A 60-year-old male presents to your office for follow-up after an upper gastrointestinal (GI) endoscopy revealed the presence of esophageal varices. His medical history is significant for cirrhosis caused by heavy alcohol abuse for the past 20 years. He was instructed to follow-up with his primary care physician for management of his condition. Which of the following is the most appropriate next step for prevention of future variceal bleeding?

Answer Choices:

A. Careful observation B. Nadolol C. Isosorbide mononitrate D. Transjugular intrahepatic portosystemic shunt

Correct Answer: B. Nadolol

Question 39

Clinical Scenario:

An 18-year-old woman presents for a routine check-up. She is a college student with no complaints. She has a 2 pack-year history of smoking and consumes alcohol occasionally. Her sexual debut was at 15 years of age and has had 2 sexual partners. She takes oral contraceptives and uses barrier contraception. Her family history is significant for cervical cancer in her aunt. Which of the following statements regarding cervical cancer screening in this patient is correct?

Answer Choices:

A. The patient requires annual Pap testing due to her family history of cervical cancer. B. The patient does not require Pap testing as long as she uses barrier contraception. C. The patient should undergo screening every 3 years after she turns 21 years of age. D. HPV testing is more preferable than Pap testing in sexually active women under 21 years of age.

Correct Answer: C. The patient should undergo screening every 3 years after she turns 21 years of age.

Question 40

Clinical Scenario:

A previously healthy 32-year-old male comes to the emergency department because of a high-grade fever and malaise for 3 days. He has severe generalized joint and body pains refractory to acetaminophen. He also has a severe stabbing pain behind his eyes. He returned from a trip to Taiwan 1 week ago. He is sexually active and uses condoms inconsistently. His temperature is 38.7°C (101.7°F), pulse is 102/min, and blood pressure is 100/70 mm Hg. Examination shows nontender inguinal lymphadenopathy. There is a maculopapular rash over the trunk and extremities with some sparing of the skin over his back and groin. Abdominal examination shows no abnormalities. Laboratory studies show: Hemoglobin 13.3 g/dL Leukocyte count 3,900/mm³ Platelet count 90,000/mm³ Serum Na⁺ 136 mEq/L Cl⁻ 103 mEq/L K⁺ 4.2 mEq/L Urea nitrogen 15 mg/dL Creatinine 1.2 mg/dL Bilirubin Total 0.4 mg/dL Direct 0.1 mg/dL Alkaline phosphatase 40 U/L AST 130 U/L ALT 60 U/L Urinalysis is normal. An ELISA test for HIV is negative. Which of the following measures is most likely to have prevented this patient's condition?"

Answer Choices:

A. Safe sexual practices B. Mosquito repellent C. Vaccination D. Frequent hand washing

Correct Answer: B. Mosquito repellent

Question 41

Clinical Scenario:

A regional academic medical center has 10 cases of adenovirus in the span of a week among its ICU patients. A committee is formed to investigate this outbreak. They are tasked with identifying the patients and interviewing the care providers to understand how adenovirus could have been spread from patient to patient. This committee will review charts, talk to the care provider teams, and investigate current patient safety and sanitation measures in the ICU. The goal of the committee is to identify weaknesses in the current system and

to put in place a plan to help prevent this sort of outbreak from reoccurring in the future. The committee is most likely using what type of analysis?

Answer Choices:

A. Failure mode and effects analysis B. Root cause analysis C. Algorithmic analysis D. Heuristic analysis

Correct Answer: B. Root cause analysis

Question 42

Clinical Scenario:

A 51-year-old man comes to the physician for 2 months of intermittent low-grade fever, malaise, and joint pain. He has a history of recurrent dental abscesses requiring drainage but has otherwise been healthy. His temperature is 38.3°C (100.9°F) and pulse is 112/min. Physical examination shows a new holosystolic murmur in the left midclavicular line that radiates to the axilla. There are linear reddish-brown lesions underneath the nail beds and tender violaceous nodules on the bilateral thenar eminences. Two sets of blood cultures grow *Streptococcus mutans*. A transthoracic echocardiogram shows moderate regurgitation of the mitral valve. Which of the following mechanisms is most likely directly involved in the pathogenesis of this patient's valvular condition?

Answer Choices:

A. Antibody cross-reaction B. Sterile platelet thrombi deposition C. Leaflet calcification and fibrosis D. Fibrin clot formation

Correct Answer: D. Fibrin clot formation

Question 43

Clinical Scenario:

A 16-year-old woman with no known past medical history and non-significant social and family histories presents to the outpatient clinic for an annual wellness checkup. She has no complaints, and her review of systems is negative. She is up to date on her childhood and adolescent vaccinations. The patient's blood pressure is 120/78 mm Hg, pulse is 82/min, respiratory rate is 16/min, and temperature is 37.0°C (98.6°F). On further questioning, she discloses that she has recently become sexual active and enquires about any necessary screening tests for cervical cancer. What is the appropriate recommendation regarding cervical cancer screening in this patient?

Answer Choices:

A. Begin 2-year interval cervical cancer screening via Pap smear today B. Begin 3-year interval cervical cancer screening via Pap smear at age 21 C. Begin 5-year interval cervical cancer screening via Pap smear at age 21 D. Offer to administer the HPV vaccine so that Pap smears can be avoided

Correct Answer: B. Begin 3-year interval cervical cancer screening via Pap smear at age 21

Question 44

Clinical Scenario:

A 6-year-old girl is brought to the emergency department because of right elbow swelling and pain 30 minutes after falling onto her outstretched right arm. She has been unable to move her right elbow since the fall. Physical examination shows bruising, swelling, and tenderness of the right elbow; range of motion is limited by pain. An x-ray of the right arm shows a supracondylar fracture of the humerus with anterior displacement of the proximal fragment. Further evaluation is most likely to show which of the following findings?

Answer Choices:

A. Absent distal radial pulse B. Radial deviation of the wrist C. Inability to abduct shoulder D. Inability to flex the elbow

Correct Answer: A. Absent distal radial pulse

Question 45

Clinical Scenario:

A 28-year-old male comes to the physician for worsening back pain. The pain began 10 months ago, is worse in the morning, and improves with activity. He has also had bilateral hip pain and difficulty bending forward during exercise for the past 3 months. He has celiac disease and eats a gluten-free diet. Examination shows a limited range of spinal flexion. Flexion, abduction, and external rotation of both hips produces pain. Further evaluation of this patient is most likely to show which of the following laboratory findings?

Answer Choices:

A. Presence of anti-dsDNA antibodies B. High levels of rheumatoid factor C. HLA-B27 positive genotype D. HLA-DR3-positive genotype

Correct Answer: C. HLA-B27 positive genotype

Question 46

Clinical Scenario:

A 25-year-old male presents to his primary care physician with a chief complaint of anxiety and fatigue. The patient states that during this past week he has had final exams and has been unable to properly study and prepare because he is so exhausted. He states that he has been going to bed early but has been unable to get a good night's sleep. The patient admits to occasional cocaine and marijuana use. Otherwise, the patient has no significant past medical history and is not taking any medications. On physical exam you note a tired and anxious appearing young man. His neurological exam is within normal limits. The patient states that he fears he will fail his courses if he does not come up with a solution. Which of the following is the best initial step in management?

Answer Choices:

A. Zolpidem B. Alprazolam C. Melatonin D. Sleep hygiene education

Correct Answer: D. Sleep hygiene education

Question 47

Clinical Scenario:

A 65-year-old woman comes to the physician for the evaluation of sharp, stabbing pain in the lower back for 3 weeks. The pain radiates to the back of her right leg and is worse at night. She reports decreased sensation around her buttocks and inner thighs. During the last several days, she has had trouble urinating. Three years ago, she was diagnosed with breast cancer and was treated with lumpectomy and radiation. Her only medication is anastrozole. Her temperature is 37°C (98.6°F), pulse is 80/min, respirations are 12/min, and blood pressure is 130/70 mm Hg. Neurologic examination shows 4/5 strength in the left lower extremity and 2/5 strength in her right lower extremity. Knee and ankle reflexes are 1+ on the right. The resting anal sphincter tone is normal but the squeeze tone is reduced. Which of the following is the most likely diagnosis?

Answer Choices:

A. Cauda equina syndrome B. Central cord syndrome C. Brown-sequard syndrome D. Anterior spinal cord syndrome

Correct Answer: A. Cauda equina syndrome

Question 48

Clinical Scenario:

A 59-year-old man presents to the health clinic for evaluation of severe itching for the past week. The itching is worse at night while lying in bed. The patient has a past medical history of hyperlipidemia, atrial fibrillation, and colon cancer. The patient takes rivaroxaban, simvastatin, and aspirin. The patient has a surgical history of colon resection, appendectomy, and tonsillectomy. He drinks a 6-pack of beer almost every night of the week. He smokes 2 packs of cigarettes daily and has been living at a homeless shelter for the past 6 months. Examination of the skin shows small crusted sores and superficial, wavy gray lines along the wrists and interdigital spaces of both hands as seen in the image. Small vesicles are also present along with excoriations. Which of the following is the most appropriate treatment option for this patient?

Answer Choices:

A. Permethrin B. Ivermectin C. Acyclovir D. Dicloxacillin

Correct Answer: A. Permethrin

Question 49

Clinical Scenario:

A 49-year-old woman presents to the clinic for a routine exam. She recently quit smoking after a 30 pack-year history and started exercising a little. Past medical history is noncontributory. She takes no medication. Her mother died at 65 from lung cancer. She rarely drinks alcohol and only uses nicotine gum as needed. She admits to having some cravings for a cigarette in the morning before work, and after work. Which of the following best describes this patient's stage in overcoming her nicotine addiction?

Answer Choices:

A. Relapse B. Maintenance C. Contemplation D. Action

Correct Answer: B. Maintenance

Question 50

Clinical Scenario:

Benzodiazepines are clinically useful because of their inhibitory effects on the central nervous system. Which of the following correctly pairs the site of action of benzodiazepines with the molecular mechanism by which they exert their effects?

Answer Choices:

A. GABA-A receptors; blocking action of GABA B. GABA-B receptors; activating a G-protein coupled receptor C. GABA-A receptors; increasing the duration of activation of a chloride ion channel D. GABA-A receptors; increasing the frequency of activation of a chloride ion channel

Correct Answer: D. GABA-A receptors; increasing the frequency of activation of a chloride ion channel

Question 51

Clinical Scenario:

A 57-year-old woman is brought to the emergency department by her husband with complaints of sudden-onset slurring for the past hour. She is also having difficulty holding things with her right hand. She denies fever, head trauma, diplopia, vertigo, walking difficulties, nausea, and vomiting. Past medical history is significant for type 2 diabetes mellitus, hypertension, and hypercholesterolemia for which she takes a baby aspirin, metformin, ramipril, and simvastatin. She has a 23-pack-year cigarette smoking history. Her blood pressure is 148/96 mm Hg, the heart rate is 84/min, and the temperature is 37.1°C (98.8°F). On physical examination, extraocular movements are intact. The patient is dysarthric, but her higher mental functions are intact. There is a right-sided facial weakness with preserved forehead wrinkling. Her gag reflex is weak. Muscle strength is mildly reduced in the right hand. She has difficulty performing skilled movements with her right hand, especially writing, and has difficulty touching far objects with her index finger. She is able to walk without difficulty. Pinprick and proprioception sensation is intact. A head CT scan is within normal limits. What is the most likely diagnosis?

Answer Choices:

A. Dysarthria-clumsy hand syndrome B. Lateral medullary syndrome C. Parinaud's syndrome D. Pure motor syndrome

Correct Answer: A. Dysarthria-clumsy hand syndrome

Question 52

Clinical Scenario:

A 42-year-old man with a history of tuberculosis presents to your office complaining of fatigue for two months. Serum laboratory studies reveal the following: WBC 7,000 cells/mm³, Hb 9.0 g/dL, Hct 25%, MCV 88 fL, Platelet 450,000 cells/mm³, Vitamin B12 500 pg/mL (200-800), and Folic acid 17 ng/mL (2.5-20). Which of the following is the most appropriate next step in the management of anemia in this patient?

Answer Choices:

A. Iron studies B. Colonoscopy C. Bone marrow biopsy D. Observation

Correct Answer: A. Iron studies

Question 53

Clinical Scenario:

A 22-year-old Caucasian G1 presents to her physician at 29 weeks gestation for a checkup. The medical history is unremarkable and the current pregnancy has been uncomplicated. Her weight is 81 kg (178.6 lb) and the height is 169 cm (5 ft 6 in). She has gained 13 kg (28.6 lb) during the pregnancy. She has no abnormalities on physical examination. Which of the following screening tests should be obtained ?

Answer Choices:

A. Non-fasting oral glucose tolerance test with 50 g of glucose B. Fasting oral glucose test with 50 g of glucose C. Non-fasting oral glucose load test with 75 g of glucose D. Measurement of HbA1c

Correct Answer: A. Non-fasting oral glucose tolerance test with 50 g of glucose

Question 54

Clinical Scenario:

A 63-year-old African American man presents to the physician for a follow-up examination. He has a history of chronic hypertension and type 2 diabetes mellitus. He has no history of coronary artery disease. His medications include aspirin, hydrochlorothiazide, losartan, and metformin. He exercises every day and follows a healthy diet. He does not smoke. He consumes alcohol moderately. There is no history of chronic disease in the family. His blood pressure is 125/75 mm Hg, which is confirmed on a repeat measurement. His BMI is 23 kg/m². The physical examination shows no abnormal findings. The laboratory test results show: Serum HbA1C 6.9% Total cholesterol 176 mg/dL Low-density lipoprotein (LDL-C) 105 mg/dL High-density

lipoprotein (HDL-C) 35 mg/dL Triglycerides 175 mg/dL The patient's 10-year risk of cardiovascular disease (CVD) is 18.7%. Lifestyle modifications including diet and exercise have been instituted. Which of the following is the most appropriate next step in pharmacotherapy?

Answer Choices:

A. Atorvastatin B. Fenofibrate C. Liraglutide D. Metoprolol

Correct Answer: A. Atorvastatin

Question 55

Clinical Scenario:

A 12-hour-old newborn is urgently transferred to the neonatal intensive care unit because he is found to be cyanotic and appears blue in all four extremities. He was born to a 42-year-old G1P1 mother who underwent no prenatal screening because she did not show up to her prenatal care visits. She has poorly controlled diabetes and hypertension. On physical exam, the infant is found to have slanted eyes, a flattened head, a large tongue, and a single palmar crease bilaterally. Furthermore, a single, loud S2 murmur is appreciated in this patient. Radiography shows a large oblong-shaped heart shadow. Based on this presentation, the neonatologist starts a medication that will temporarily ensure this patient's survival pending definitive fixation of the defect. The substance that was most likely administered in this case also has which of the following effects?

Answer Choices:

A. Decreased platelet aggregation B. Increased bronchial tone C. Vascular vasoconstriction D. Vascular vasodilation

Correct Answer: D. Vascular vasodilation

Question 56

Clinical Scenario:

A 67-year-old woman is brought to the emergency department for the evaluation of fever, chest pain, and a cough productive of a moderate amount of greenish-yellow sputum for 2 days. During this period, she has had severe malaise, chills, and difficulty breathing. She has hypertension, hypercholesterolemia, and type 2 diabetes mellitus. She smoked one pack of cigarettes daily for 20 years, but quit 5 years ago. Current medications include simvastatin, captopril, and metformin. Temperature is 39°C (102.2°F), pulse is 110/min, respirations are 33/min, and blood pressure is 143/88 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 94%. Crackles are heard on auscultation of the right upper lobe. Laboratory studies show a leukocyte count of 12,300/mm³, an erythrocyte sedimentation rate of 60 mm/h, and a urea nitrogen of 15 mg/dL. A chest x-ray is shown. Which of the following is the most appropriate next step in the management of this patient?

Answer Choices:

A. Inpatient treatment with cefepime, azithromycin, and gentamicin B. Outpatient treatment with azithromycin and amoxicillin-clavulanate C. Inpatient treatment with azithromycin and cefotaxime D. Inpatient treatment with ceftriaxone

Correct Answer: C. Inpatient treatment with azithromycin and cefotaxime

Question 57

Clinical Scenario:

In a healthy patient with no renal abnormalities, several mechanisms are responsible for moving various filtered substances into and out of the tubules. Para-aminohippurate (PAH) is frequently used to estimate renal blood flow when maintained at low plasma concentrations. The following table illustrates the effect of changing plasma PAH concentrations on PAH excretion: Plasma PAH concentration (mg/dL) Urinary PAH concentration (mg/dL) 0 0 10 60 20 120 30 150 40 180 Which of the following mechanisms best explains the decrease in PAH excretion with the increase in plasma concentration greater than 20 mg/dL?

Answer Choices:

A. Saturation of PAH transport carriers B. Increased diffusion rate of PAH C. Decreased glomerular filtration of PAH D. Increased flow rate of tubular contents

Correct Answer: A. Saturation of PAH transport carriers

Question 58

Clinical Scenario:

A 33-year-old woman comes to the physician because of a 6-month history of worsening shortness of breath and fatigue. Her paternal uncle had similar symptoms and died of respiratory failure at 45 years of age. The lungs are clear to auscultation. Pulmonary function testing shows an FVC of 84%, an FEV1/FVC ratio of 92%, and a normal diffusion capacity. An ECG shows a QRS axis greater than +90 degrees. Genetic analysis shows an inactivating mutation in the bone morphogenetic protein receptor type II (BMPRII) gene. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. Chronic intravascular hemolysis B. Fibrosis of the pulmonary parenchyma C. Thickening of the interventricular septum D. Elevated pulmonary arterial pressure

Correct Answer: D. Elevated pulmonary arterial pressure

Question 59

Clinical Scenario:

A 66-year-old farmer is being evaluated for abnormal lung findings on a low dose chest CT scan obtained as part of his lung cancer screening. He has a 50-pack-year smoking history and has been hesitant to quit. He has a non-productive cough but brushes it away saying he is not bothered by it. He denies ever coughing up blood, breathlessness, chest pain, fatigue, or weight loss. He has never sought any medical care and states that he has always been in good shape. He consumes alcohol moderately and uses marijuana occasionally. He lives with his wife and has not traveled recently. On physical examination, his temperature is 37.1°C (98.8°F), blood pressure is 148/70 mm Hg, and pulse rate is 95/min. His BMI is 32 kg/m². A general physical examination is unremarkable. Coarse breath sounds are present bilaterally. The cardiac exam is normal. Laboratory studies show a normal complete blood count and comprehensive metabolic panel. A follow-up high-resolution CT scan is performed that shows small irregular subcentimeter pulmonary nodules, several of which are cavitated in both lungs, predominantly distributed in the upper and middle zones. There is no mediastinal or hilar lymphadenopathy. A transbronchial needle aspiration of the lesion is performed which shows a nodular pattern of abundant, granular, mildly eosinophilic cells with grooved nuclei with indented nuclear membranes and a chronic inflammation that consists primarily of eosinophils. Immunohistochemical staining reveals numerous cells that stain positive for S100 and CD1a. Which of the cells of the human immune system are responsible for this lesion?

Answer Choices:

A. T lymphocytes B. Natural killer cells C. Dendritic cells D. Ciliary epithelium

Correct Answer: C. Dendritic cells

Question 60

Clinical Scenario:

A 54-year-old woman presents to the physician with discomfort in her upper left abdomen over the past month. Moreover, she has recently been feeling a bit tired. She has no history of any significant illness and takes no medications. Her vital signs are within normal limits. On percussion, the spleen size is 15 cm (5.9 in). Otherwise, the physical examination shows no abnormalities. The laboratory test results are as follows: Hemoglobin 10 g/dL Mean corpuscular volume 88 µm³ Leukocyte count 65,000/mm³ Platelet count 500,000/mm³ Two images of the peripheral blood smear are shown. Laboratory studies are most likely to show which of the following?

Answer Choices:

A. Auer rods B. JAK2 mutation C. Philadelphia chromosome D. Translocation between chromosomes 15 and 17

Correct Answer: C. Philadelphia chromosome

Question 61

Clinical Scenario:

A 43-year-old man presents with acute-onset left flank pain for the past 6 hours. He describes the pain as severe, intermittent, colicky, and “coming in waves”, and he points to the area of the left costovertebral angle (CVA). He says he recently has been restricting oral liquid intake to only 2 glasses of water per day based on the advice of his healer. He also reports nausea and vomiting. The patient has a history of hypertension, gout, and type 2 diabetes mellitus. He is afebrile, and his vital signs are within normal limits. On physical examination, he is writhing in pain and moaning. There is exquisite left CVA tenderness. A urinalysis shows gross hematuria. Which of the following is the next best step in the management of this patient?

Answer Choices:

A. Contrast CT of the abdomen and pelvis B. Renal ultrasound C. Non-contrast CT of the abdomen and pelvis D. Supine abdominal radiograph

Correct Answer: C. Non-contrast CT of the abdomen and pelvis

Question 62

Clinical Scenario:

An 11-year-old girl presents to her pediatrician for evaluation of asymmetry that was detected during routine school screening. Specifically, she was asked to bend forwards while the school nurse examined her back. While leaning forward, her right scapula was found to be higher than her left scapula. She was also found to have a prominent line of spinal processes that diverged from the midline. She has been experiencing some back pain that she previously attributed to growth pains but otherwise has no symptoms. Her past medical history is significant only for mild allergies. She is sent to radiography for confirmation of the diagnosis and placed in a nighttime brace. Which of the following represents a complication of the most likely disease affecting this patient if it is left untreated?

Answer Choices:

A. Congestive heart failure B. Dislocation of the shoulders C. Obstructive lung disease D. Restrictive lung disease

Correct Answer: D. Restrictive lung disease

Question 63

Clinical Scenario:

A 23-year-old woman presents with fever, chills, nausea, and urinary urgency and frequency. She says that her symptoms began 4 days ago and have progressively worsened. Her past medical history is significant for a 6-month history of recurrent urinary tract infections (UTIs). Her vital signs include: temperature 39.0°C (102.2°F), blood pressure 100/70 mm Hg, pulse 92/min, and respiratory rate 25/min. On physical examination, there is moderate left costovertebral angle tenderness. Laboratory findings are significant for the following: WBC 8,500/mm³ RBC 4.20 x 10⁶/mm³ Hematocrit 41.5% Hemoglobin 13.0 g/dL Platelet count 225,000/mm³ Urinalysis Color Dark yellow Clarity Turbid pH 6.5 Specific gravity 1.026 Glucose None Ketones None

Nitrites Positive Leukocyte esterase Positive Bilirubin Negative Urobilirubin 0.6 mg/dL Protein Trace Blood None WBC 25/hpf Bacteria Many Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Pyelonephritis B. Uncomplicated cystitis C. UTI D. Acute obstructing nephrolithiasis

Correct Answer: A. Pyelonephritis

Question 64

Clinical Scenario:

A 38-year-old woman, gravida 3, para 2, at 12 weeks' gestation comes to her obstetrician for a prenatal visit. Screening tests in the first trimester showed a decreased level of pregnancy-associated plasma protein and an increased level of β -hCG. A genetic disorder is suspected. Which of the following results from an additional diagnostic test is most likely to confirm the diagnosis?

Answer Choices:

A. Additional chromosome in placental tissue B. Decreased estriol in maternal serum C. Triploidy in amniotic fluid D. Decreased inhibin A in maternal serum

Correct Answer: A. Additional chromosome in placental tissue

Question 65

Clinical Scenario:

E. coli has the ability to regulate its enzymes to break down various sources of energy when available. It prevents waste by the use of the lac operon, which encodes a polycistronic transcript. At a low concentration of glucose and absence of lactose, which of the following occurs?

Answer Choices:

A. Increased cAMP levels result in binding to the catabolite activator protein B. Decreased cAMP levels result in poor binding to the catabolite activator protein C. Transcription of the lac Z, Y, and A genes increase D. Repressor releases from lac operator

Correct Answer: A. Increased cAMP levels result in binding to the catabolite activator protein

Question 66

Clinical Scenario:

A 49-year-old woman presents to her physician with a fever accompanied by chills and burning micturition since the past 5 days. She is an otherwise healthy woman with no significant past medical history and has an active sexual life. On physical examination, her temperature is 39.4°C (103.0°F), pulse rate is 90/min, blood

pressure is 122/80 mm Hg, and respiratory rate is 14/min. Examination of the abdomen and genitourinary region do not reveal any specific positive findings. The physician orders a urinalysis of fresh unspun urine for this patient which shows 25 WBCs/mL of urine. The physician prescribes an empirical antibiotic and other medications for symptom relief. He also orders a bacteriological culture of her urine. After 48 hours of treatment, the woman returns to the physician to report that her symptoms have not improved. The bacteriological culture report indicates the growth of gram-negative bacilli which are lactose-negative and indole-negative, which produce a substance that hydrolyzes urea to produce ammonia. Which of the following bacteria is the most likely cause of infection in the woman?

Answer Choices:

A. *Citrobacter freundii* B. *Escherichia coli* C. *Enterobacter cloacae* D. *Proteus mirabilis*

Correct Answer: D. *Proteus mirabilis*

Question 67

Clinical Scenario:

A 49-year-old woman comes to the physician because of difficulty walking and dizziness for the past 2 weeks. She has also had fatigue, heartburn, and diarrhea for 4 months. The stools are foul-smelling and do not flush easily. Over the past 4 months, she has had a 2.2-kg (5-lb) weight loss. Her only medication is an over-the-counter antacid. Her mother has autoimmune thyroid disease and Crohn disease. She is 150 cm (4 ft 11 in) tall and weighs 43 kg (95 lb); BMI is 19.1 kg/m². Vital signs are within normal limits. Examination shows a wide-based gait. Muscle strength and tone are normal in all extremities. Rapid alternating movement of the hands is impaired. The abdomen is soft and there is mild tenderness to palpation in the epigastric area. Her hemoglobin concentration is 11.1 mg/dL, and levels of vitamin E and vitamin D are decreased. Upper endoscopy shows several ulcers in the gastric antrum and the descending duodenum. Which of the following is the most likely underlying mechanism of this patient's symptoms?

Answer Choices:

A. Inactivation of pancreatic enzymes B. *T. whipplei* infiltration of intestinal villi C. Intestinal inflammatory reaction to gluten D. Small intestine bacterial overgrowth

Correct Answer: A. Inactivation of pancreatic enzymes

Question 68

Clinical Scenario:

A 37-year-old patient is being evaluated for involuntary movements, difficulty swallowing food, and personality change. He has entered a clinical trial that is studying the interaction of certain neuromediators in patients with similar (CAG) n trinucleotide repeat disorders. The laboratory results of 1 of the candidates for the clinical trial are presented below: Acetylcholine ↓ Dopamine ↑ Gamma-aminobutyric acid (GABA) ↓ Norepinephrine unchanged Serotonin unchanged Which trinucleotide disorder most likely represents the diagnosis of this patient?

Answer Choices:

A. Myotonic dystrophy B. Friedreich's ataxia C. Fragile X syndrome D. Huntington's disease

Correct Answer: D. Huntington's disease

Question 69**Clinical Scenario:**

A 28-year-old woman presents with a 12-month history of headache, tinnitus, retrobulbar pain, and photopsias. She says the headaches are mild to moderate, intermittent, diffusely localized, and refractory to nonsteroidal anti-inflammatory drugs (NSAIDs). In addition, this past week, she began to have associated dizziness and photopsia with the headaches. Physical examination reveals a body temperature of 36.5°C (97.7°F), blood pressure of 140/80 mm Hg, and a respiratory rate of 13/min and regular. BMI is 29 kg/m². Neurological examination is significant for peripheral visual field loss in the inferior nasal quadrant, diplopia, bilateral abducens nerve palsy, and papilledema. A T1/T2 MRI of the brain did not identify extra-axial or intra-axial masses or interstitial edema, and a lumbar puncture showed an opening pressure of 27 cm H₂O, with a cerebrospinal fluid analysis within the normal range. Which of the following best describes the pathogenic mechanism underlying these findings?

Answer Choices:

A. Arachnoid granulation adhesions B. Elevated intracranial venous pressure C. Increased cerebrospinal production D. Systemic hypertension

Correct Answer: B. Elevated intracranial venous pressure

Question 70**Clinical Scenario:**

A 46-year-old premenopausal woman undergoes lumpectomy after a diagnosis of invasive ductal carcinoma of the breast is made. Pathologic examination of the surgical specimen shows that the breast cancer cells stain positive for estrogen receptor and progesterone receptor, and negative for human epidermal growth factor receptor 2. Which of the following characteristics applies to the most appropriate pharmacotherapy for this patient's condition?

Answer Choices:

A. Selective agonist at estrogen receptors in bone tissue B. Monoclonal antibody against tyrosine kinase receptor C. Monoclonal antibody against vascular endothelial growth factor D. Selective antagonist at estrogen receptors in endometrium "

Correct Answer: A. Selective agonist at estrogen receptors in bone tissue

Question 71

Clinical Scenario:

A 2900-g (6.4-lb) male newborn is delivered at term to a 29-year-old primigravid woman. His mother had no routine prenatal care. She reports that the pregnancy was uncomplicated apart from a 2-week episode of a low-grade fever and swollen lymph nodes during her early pregnancy. She has avoided all routine vaccinations because she believes that "natural immunity is better." The newborn is at the 35th percentile for height, 15th percentile for weight, and 3rd percentile for head circumference. Fundoscopic examination shows inflammation of the choroid and the retina in both eyes. A CT scan of the head shows diffuse intracranial calcifications and mild ventriculomegaly. Prenatal avoidance of which of the following would have most likely prevented this newborn's condition?

Answer Choices:

A. Exposure to unvaccinated children B. Undercooked pork C. Unprotected sexual intercourse D. Mosquito bites

Correct Answer: B. Undercooked pork

Question 72

Clinical Scenario:

A 9-year-old boy presents for incision and drainage of a small abscess on his left thigh. No significant past medical history. No current medications. Before the procedure, the patient is allowed to inhale colorless, sweet-smelling gas. After the procedure, the patient receives 3–4 minutes of high flow oxygen through a nasal mask. The pulse oximetry shows an oxygen saturation of 94%. This patient was oxygenated at the end of the procedure to prevent which of the following complications?

Answer Choices:

A. Cardiac arrhythmias B. Raised intracranial pressure C. Hepatotoxicity D. Diffusion hypoxia

Correct Answer: D. Diffusion hypoxia

Question 73

Clinical Scenario:

A 20-year-old male is brought by ambulance to the emergency room in extremis. He is minimally conscious, hypotensive, and tachypneic, and his breath gives off a "fruity" odor. An arterial blood gas and metabolic panel show anion gap metabolic acidosis. This patient is most likely deficient in which of the following metabolic actions?

Answer Choices:

A. Formation of ketone bodies B. Glucose production C. Cortisol secretion D. Cellular uptake of glucose

Correct Answer: D. Cellular uptake of glucose

Question 74

Clinical Scenario:

A 23-year-old woman comes to the physician because of a 5-month history of a pruritic rash on the bilateral upper extremities. She has no history of serious illness and takes no medications. A skin biopsy of the rash shows intraepidermal accumulation of edematous fluid and widening of intercellular spaces between keratinocytes. Which of the following is the most likely diagnosis?

Answer Choices:

A. Psoriasis vulgaris B. Lichen planus C. Acanthosis nigricans D. Eczematous dermatitis

Correct Answer: D. Eczematous dermatitis

Question 75

Clinical Scenario:

One day after a 4700-g (10-lb 6-oz) male newborn is delivered to a 28-year-old primigravid woman, the newborn has bluish discoloration of the lips and fingernails. His temperature is 37.3°C (99.1°F), pulse is 166/min, respirations are 63/min, and blood pressure is 68/44 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 81%. Examination shows central cyanosis. A grade 2/6 holosystolic murmur is heard over the left lower sternal border. A single second heart sound is present. Supplemental oxygen does not improve cyanosis. An x-ray of the chest shows an enlarged cardiac silhouette with a narrowed mediastinum. Further evaluation of the mother is most likely to show which of the following?

Answer Choices:

A. Increased serum TSH B. Prenatal lithium intake C. Positive rapid plasma reagin test D. Elevated hemoglobin A1c

Correct Answer: D. Elevated hemoglobin A1c

Question 76

Clinical Scenario:

A 24-year-old man is brought to the emergency department 30 minutes after being involved in a high-speed motor vehicle collision in which he was a restrained driver. On arrival, he is alert and oriented. His pulse is 112/min, respirations are 29/min, and blood pressure is 100/60 mm Hg. The pupils are equal and reactive to light. There is a 3-cm laceration over the forehead and multiple bruises over the trunk. The lungs are clear to auscultation. Cardiac examination shows no abnormalities. The abdomen is soft and nontender. The right knee is swollen and tender; range of motion is limited by pain. Infusion of 0.9% saline is begun and intravenous

acetaminophen is administered. Two hours later, blood-tinged fluid spontaneously drains from both nostrils, and is made worse by leaning forward. On a piece of gauze, it shows up as a rapidly-expanding clear ring of fluid surrounding blood. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Cranial nerve XII palsy B. Bilateral periorbital ecchymosis C. Numbness of upper cheek area D. Retroauricular ecchymosis "

Correct Answer: B. Bilateral periorbital ecchymosis

Question 77

Clinical Scenario:

A 66-year-old woman comes to the physician because of a 1-week history of pruritic blister formation. Physical examination shows multiple 1–3 cm bullae on the palms, soles, lower legs, and inguinal folds. Gentle rubbing of the skin does not result in sloughing of the epidermis. Immunofluorescence studies of a perilesional skin biopsy specimen are most likely to show deposition of antibodies in which of the following areas?

Answer Choices:

A. At the dermoepidermal junction B. Between epidermal keratinocytes C. In dermal papillae D. In dermal vessel walls

Correct Answer: A. At the dermoepidermal junction

Question 78

Clinical Scenario:

A 70 year-old-man comes to the clinic for generalized fatigue. He says that he is more tired than before and has difficulty catching his breath while walking upstairs. He feels tired quickly doing his usual activity such as gardening and shopping. He does not have any symptoms of fever, change in bowel habits, abdominal pain, rectal bleeding, or weight loss. His appetite is normal. His last colonoscopy was done 10 years ago and it was normal. His blood pressure is 116/74 and heart rate is 87/min. On physical examination, his conjunctivae are pale. A routine blood test shows iron deficiency anemia with hemoglobin of 10 gm/dL. His stool is positive for occult blood. He is then sent for a colonoscopy (image is shown). What is the most likely diagnosis for the above condition?

Answer Choices:

A. Diverticulitis B. Hereditary hemorrhagic telangiectasia C. Angiodysplasia D. Intestinal obstruction

Correct Answer: C. Angiodysplasia

Question 79

Clinical Scenario:

A 4-year-old girl presents to the office with her parents who are concerned about their daughter and slow, progressive changes in her behavior. The girl was born at 39 weeks gestation via spontaneous vaginal delivery. She is up to date on all vaccines and was meeting all developmental milestones until about 2 years ago. At one point she had a vocabulary of several words and now she verbalizes in grunts. She also flaps her hands in a repeated motion and has difficulty walking. Her parents have tried several home therapies to improve their daughter's symptoms including restricted diets, hydrotherapy, and a variety of nutritional supplements. The vital signs include: heart rate 90/min, respiratory rate 22/min, blood pressure 110/65 mm Hg, and temperature 36.9°C (98.4°F). On physical exam, she is well nourished and stares absently out the window. Her heart has a regular rate and rhythm and her lungs are clear to auscultation bilaterally. She has mild scoliosis. Which of the following is the most likely diagnosis?

Answer Choices:

A. Autistic spectrum disorder B. Phenylketonuria C. Rett syndrome D. Tourette syndrome

Correct Answer: C. Rett syndrome

Question 80

Clinical Scenario:

A 68-year-old man comes to the physician because of double vision and unilateral right eye pain that began this morning. His vision improves when he covers either eye. He has hypertension, mild cognitive impairment, and type 2 diabetes mellitus. The patient has smoked two packs of cigarettes daily for 40 years. His current medications include lisinopril, donepezil, metformin, and insulin with meals. His temperature is 37°C (98.6°F), pulse is 85/minute, respirations are 12/minute, and blood pressure is 132/75 mm Hg. His right eye is abducted and depressed with slight intorsion. He can only minimally adduct the right eye. Visual acuity is 20/20 in both eyes. Extraocular movements of the left eye are normal. An MRI of the head shows no abnormalities. His fingerstick blood glucose concentration is 325 mg/dL. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Ptosis B. Dilated and fixed pupil C. Miosis and anhidrosis D. Positive swinging-flashlight test "

Correct Answer: A. Ptosis

Question 81

Clinical Scenario:

A 15-year-old girl comes to the physician with her father for evaluation of short stature. She feels well overall but is concerned because all of her friends are taller than her. Her birth weight was normal. Menarche has not

yet occurred. Her father says he also had short stature and late puberty. The girl is at the 5th percentile for height and 35th percentile for weight. Breast development is Tanner stage 2. Pubic and axillary hair is absent. An x-ray of the left hand and wrist shows a bone age of 12 years. Further evaluation of this patient is most likely to show which of the following sets of laboratory findings? \$\$\$ FSH %%% LH %%% Estrogen %%% GnRH \$\$\$

Answer Choices:

A. ↓ ↓ ↓ ↓ B. ↓ ↓ ↑ ↓ C. Normal normal normal normal D. ↑ ↑ ↓ ↑

Correct Answer: C. Normal normal normal normal

Question 82

Clinical Scenario:

A 25-year-old man comes to the physician because of a 2-week history of numbness in his left lower extremity. One month ago, he sustained a fracture of the neck of the left fibula during soccer practice that was treated with immobilization in a plaster cast. Physical examination of the left lower extremity is most likely to show which of the following findings?

Answer Choices:

A. Impaired dorsiflexion of the foot B. Loss of sensation over the medial calf C. Inability to stand on tiptoes D. Decreased ankle reflex

Correct Answer: A. Impaired dorsiflexion of the foot

Question 83

Clinical Scenario:

A 42-year-old woman presents with exertional dyspnea and fatigue for the past 3 months. Her past medical history is significant for multiple episodes of mild diarrhea for many years, which was earlier diagnosed as irritable bowel syndrome (IBS). She denies any current significant gastrointestinal symptoms. The patient is afebrile and vital signs are within normal limits. Physical examination reveals oral aphthous ulcers and mild conjunctival pallor. Abdominal examination is unremarkable. There is a rash present on the peripheral extremities bilaterally (see image). Laboratory findings are significant for evidence of microcytic hypochromic anemia. FOBT is negative. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Small intestinal bacterial overgrowth B. Non-tropical sprue C. Whipple's disease D. Irritable bowel disease

Correct Answer: B. Non-tropical sprue

Question 84

Clinical Scenario:

2 hours after being admitted to the hospital because of a fracture of the right ankle, a 75-year-old man continues to complain of pain despite treatment with acetaminophen and ibuprofen. He has a history of dementia and cannot recall his medical history. The presence of which of the following features would most likely be a reason to avoid treatment with morphine in this patient?

Answer Choices:

A. Watery diarrhea B. Biliary tract dysfunction C. Persistent cough D. Tachypnea

Correct Answer: B. Biliary tract dysfunction

Question 85

Clinical Scenario:

A 14-year-old boy is brought to the office by his step-parents because he was recently caught beating a stray cat in an alley near his home. He has a police record which includes vandalism, shoplifting, and running away on two occasions. He has also received several detentions and threats of expulsion from school due to bullying and being too aggressive with the younger students. Past medical history is significant for a history of ADHD previously treated with methylphenidate, but now he does not take anything. His biological family placed him and his sister into the foster care system. His step-parents try to provide support and nurturing home life but the patient is very resistant and often acts out. What is the most likely diagnosis for this patient?

Answer Choices:

A. Antisocial personality disorder B. Attention deficit hyperactivity disorder C. Conduct disorder D. Schizoid personality disorder

Correct Answer: C. Conduct disorder

Question 86

Clinical Scenario:

A 36-year-old man presents to his primary care physician because of shortness of breath. He is an office worker who has a mostly sedentary lifestyle; however, he has noticed that recently he feels tired and short of breath when going on long walks with his wife. He also has had a hacking cough that seems to linger, though he attributes this to an upper respiratory tract infection he had 2 months ago. He has diabetes that is well-controlled on metformin and has smoked 1 pack per day for 20 years. Physical exam reveals a large chested man with wheezing bilaterally and mild swelling in his legs and abdomen. The cause of this patient's abdominal and lower extremity swelling is most likely due to which of the following processes?

Answer Choices:

A. Damage to kidney tubules B. Defective protein folding C. Excessive protease activity D. Hyperplasia of mucous glands

Correct Answer: B. Defective protein folding

Question 87

Clinical Scenario:

A 61-year-old man comes to the physician because of a 3-month history of fatigue and progressively worsening shortness of breath that is worse when lying down. Recently, he started using two pillows to avoid waking up short of breath at night. Examination shows a heart murmur. A graph with the results of cardiac catheterization is shown. Given this patient's valvular condition, which of the following murmurs is most likely to be heard on cardiac auscultation?

Answer Choices:

A. High-pitched, holosystolic murmur that radiates to the axilla B. Rumbling, delayed diastolic murmur heard best at the cardiac apex C. Blowing, early diastolic murmur heard best at the Erb point D. Harsh, late systolic murmur that radiates to the carotids

Correct Answer: A. High-pitched, holosystolic murmur that radiates to the axilla

Question 88

Clinical Scenario:

A 1-year-old immigrant girl presents to her pediatrician for a routine well-child check. She has not received any recommended vaccines since birth. She attends daycare and remains healthy despite her daily association with several other children for the past 3 months at a home daycare facility. Which of the following phenomena explains why she has not contracted any vaccine-preventable diseases such as measles, diphtheria, or pertussis?

Answer Choices:

A. Herd immunity B. Genetic drift C. Tolerance D. Immune evasion

Correct Answer: A. Herd immunity

Question 89

Clinical Scenario:

A 54-year-old man is brought to the emergency department 30 minutes after being hit by a car while crossing the street. He had a left-sided tonic-clonic seizure and one episode of vomiting while being transported to the hospital. On arrival, he is not oriented to person, place, or time. Physical examination shows flaccid paralysis of all extremities. A CT scan of the head is shown. This patient's symptoms are most likely the result of a hemorrhage in which of the following structures?

Answer Choices:

A. Into the ventricular system B. Between the skull and the dura mater C. Into the cerebral parenchyma D. Between the arachnoid mater and the pia mater

Correct Answer: D. Between the arachnoid mater and the pia mater

Question 90**Clinical Scenario:**

A scientist is studying patients with neuromuscular weakness and discovers a mutation in a plasma membrane ion channel. She thinks that this mutation may have an effect on the dynamics of action potentials so she investigates its effect in an isolated neuronal membrane. She finds that the ion channel has no effect when potassium, sodium, and calcium are placed at physiological concentrations on both sides of the membrane; however, when some additional potassium is placed inside the membrane, the channel rapidly allows for sodium to enter the membrane. She continues to examine the mutant channel and finds that it is more rapidly inactivated compared with the wildtype channel. Which of the following effects would this mutant channel most likely have on the electrical profile of neurons in these patients?

Answer Choices:

A. Decreased action potential amplitude B. Decreased hyperpolarization potential C. Decreased resting membrane potential D. Increased action potential refractory period

Correct Answer: A. Decreased action potential amplitude

Question 91**Clinical Scenario:**

A 35-year-old man is brought to the emergency department by his wife because of a 1-week history of progressive confusion, myalgia, and nausea. His wife says that he first reported headaches and fatigue 10 days ago, and since then “he has not been himself”. He has refused to drink any liquids for the last day. Two months ago, he helped his neighbor remove a raccoon’s den from her backyard. He appears agitated. His temperature is 100.8°F (38.2°C). Examination shows excessive drooling. Muscle tone and deep tendon reflexes are increased bilaterally. Administration of which of the following is most likely to have prevented this patient’s condition?

Answer Choices:

A. RNA-dependent DNA polymerase inhibitor B. Chemically-inactivated virus C. Live attenuated vaccine D. Immunoglobulin against a bacterial protein

Correct Answer: B. Chemically-inactivated virus

Question 92

Clinical Scenario:

A 45-year-old obese man is evaluated in a locked psychiatric facility. He was admitted to the unit after he was caught running through traffic naked while tearing out his hair. His urine toxicology screening was negative for illicit substances and after careful evaluation and additional history, provided by his parents, he was diagnosed with schizophrenia and was treated with aripiprazole. His symptoms did not improve after several dosage adjustments and he was placed on haloperidol, but this left him too lethargic and slow and he was placed on loxapine. After several dosage adjustments today, he is still quite confused. He describes giant spiders and robots that torture him in his room. He describes an incessant voice screaming at him to run away. He also strongly dislikes his current medication and would like to try something else. Which of the following is indicated in this patient?

Answer Choices:

A. Fluphenazine B. Haloperidol C. Olanzapine D. Clozapine

Correct Answer: D. Clozapine

Question 93

Clinical Scenario:

A 25-year-old woman presents to the ED with a diffuse, erythematous rash in the setting of nausea, vomiting, and fever for 2 days. Physical exam reveals a soaked tampon in her vagina. Blood cultures are negative. The likely cause of this patient's disease binds to which molecule on T cells?

Answer Choices:

A. CD3 B. CD40 ligand C. Variable beta portion of the T-cell receptor D. Fas ligand

Correct Answer: C. Variable beta portion of the T-cell receptor

Question 94

Clinical Scenario:

A 56-year-old man comes to the physician because of intermittent retrosternal chest pain. Physical examination shows no abnormalities. Endoscopy shows salmon pink mucosa extending 5 cm proximal to the gastroesophageal junction. Biopsy specimens from the distal esophagus show nonciliated columnar epithelium with numerous goblet cells. Which of the following is the most likely cause of this patient's condition?

Answer Choices:

A. Neoplastic proliferation of esophageal epithelium B. Esophageal exposure to gastric acid C. Hypermotile esophageal contractions D. Fungal infection of the lower esophagus

Correct Answer: B. Esophageal exposure to gastric acid

Question 95

Clinical Scenario:

A 28-year-old pregnant female presents for a prenatal check-up at 20 weeks gestation, which includes routine screening ultrasound. Fetal ultrasound demonstrates bilateral multicystic dysplastic kidneys. Her pregnancy has been complicated by persistent oligohydramnios. The child requires significant pulmonary support upon delivery. Which of the following clinical findings is most likely present in this child as a result of these abnormalities?

Answer Choices:

A. Urachal fistula B. Esophageal atresia C. Spina bifida occulta D. Clubbed feet

Correct Answer: D. Clubbed feet

Question 96

Clinical Scenario:

A 31-year-old man presents with a several-month history of foot sensory changes. He has noticed that he has a hard time telling the difference between a hardwood floor and carpet beneath his feet. He's also had a couple of falls lately; these falls were not preceded by any lightheadedness or palpitations. He is adopted, so his family history is unknown. On physical exam, he has leg and foot muscular atrophy and 4/5 strength throughout his bilateral lower extremities. Sensation to light touch and pinprick is decreased up to the mid-calf. Ankle jerk reflex is absent bilaterally. He has a significant pes cavus deformity of both feet. Nerve conduction studies show decreased conduction velocities in his bilateral peroneal nerves. Which of the following is the best treatment for this patient?

Answer Choices:

A. Aspirin B. Cilostazol C. Referral to orthopedic surgery D. Referral to physical therapy

Correct Answer: D. Referral to physical therapy

Question 97

Clinical Scenario:

A 31-year-old woman is brought to the emergency department because of a severe right-sided temporal headache with conjunctival swelling and anterior bulging of the left eye for 1 hour. The patient has had right-sided purulent nasal discharge and nasal congestion for the past 2 days. There is no personal or family history of serious illness. The patient does not smoke or drink alcohol. She takes no medications. She appears to be in acute distress. Her temperature is 40°C (104°F), pulse is 105/min, and blood pressure is 125/80 mm Hg. Examination shows bilateral ptosis. The pupils are equal and reactive to light; lateral gaze of the left eye is

limited. Ophthalmic examination shows periorbital edema and chemosis of the left eye. The remainder of the examination shows no abnormalities. The patient most likely requires treatment with which of the following?

Answer Choices:

- A. Intranasal sumatriptan B. Surgical debridement C. Intravenous vancomycin, ceftriaxone, and metronidazole
D. Intravenous dihydroergotamine "

Correct Answer: C. Intravenous vancomycin, ceftriaxone, and metronidazole

Question 98

Clinical Scenario:

A 67-year-old woman who was recently diagnosed with Crohn disease comes to the physician for evaluation of her immunosuppressive therapy. She has had recurrent flares since her diagnosis. Physical examination shows two shallow ulcers on her oral mucosa. The physician considers adding azathioprine to her medication regimen. A deficiency of which of the following enzymes would diminish the therapeutic effect of this drug?

Answer Choices:

- A. Xanthine oxidase B. Thymidylate synthase C. Phosphoribosyl pyrophosphate synthetase D. Hypoxanthine-guanine phosphoribosyl transferase

Correct Answer: D. Hypoxanthine-guanine phosphoribosyl transferase

Question 99

Clinical Scenario:

A 25-year-old woman comes to the physician because she has noted darkening of the skin around her neck since wearing a chain she recently bought at a thrift shop. The darkening occurred gradually over the past 2 months and is accompanied by thickening of the affected skin. She has peptic ulcer disease. Menses occur at irregular 35- to 60-day intervals and last for 9 days with heavy flow. Menarche was at the age of 14 years and her last menstrual period was 3 weeks ago. She is sexually active with her husband and they do not use contraception. The patient's only medication is cimetidine. She is 163 cm (5 ft 4 in) tall and weighs 91 kg (200 lb); BMI is 34 kg/m². Vital signs are within normal limits. Physical examination shows velvety, hyperpigmented plaques in the axillae, the inframammary fold, and around the neck. The remainder of the examination shows no abnormalities. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

- A. Diffusely enlarged thyroid gland on ultrasonography of the neck B. Elevated serum 17-hydroxyprogesterone levels C. Malignant glandular cells on gastric biopsy D. Polycystic ovaries on ultrasonography of the pelvis

Correct Answer: D. Polycystic ovaries on ultrasonography of the pelvis

Question 100

Clinical Scenario:

A 75-year-old man is brought to the emergency room after being found unconscious in his home. His medical history is unknown. On physical examination he does not demonstrate any spontaneous movement of his extremities and is unable to respond to voice or painful stimuli. You notice that he is able blink and move his eyes in the vertical plane. Based on these physical exam findings, you expect that magnetic resonance angiogram will most likely reveal an occlusion in which of the following vessels?

Answer Choices:

A. Anterior cerebral artery B. Anterior inferior cerebellar artery C. Basilar artery D. Posterior cerebral artery

Correct Answer: C. Basilar artery

Question 101

Clinical Scenario:

A 16-year-old girl is brought to the physician by her mother because she has not attained menarche. She has no history of serious illness. She is at 50th percentile for height and weight. Examination shows no breast glandular tissue and no pubic hair development. The remainder of the examination shows no abnormalities. A urine pregnancy test is negative. An ultrasound of the pelvis shows no abnormalities. Which of the following is the most appropriate next step in management?

Answer Choices:

A. GnRH stimulation test B. Reassurance C. Progesterone challenge test D. Serum FSH level

Correct Answer: D. Serum FSH level

Question 102

Clinical Scenario:

A 25-year-old woman, gravida 2, para 1, is brought to the emergency department at 39 weeks' gestation in active labor. Her first child was delivered at 40 weeks' gestation by elective cesarean section due to limited range of motion in her hip. The patient has sickle cell disease. She has had multiple episodes of acute chest syndrome and has required several transfusions in the past. An uncomplicated repeat cesarean section is performed, and a 2.7-kg (6-lb) infant is delivered with approx. 550 mL blood loss. Perioperatively, she received one dose of intravenous cefazolin. Following the surgery, the patient continues to bleed, and she receives a transfusion of 1 unit of packed red blood cells. One hour later, the patient begins to have flank pain and appears to be in acute distress. Her temperature is 38.5°C (101.3°F), pulse is 111/min, respirations are 22/min, and blood pressure is 99/50 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 97%. Foley catheter shows dark brown urine. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Serum antibodies against Class I HLA antigens B. Positive blood cultures C. Positive direct Coombs test D. Bilateral pulmonary infiltrates on chest x-ray

Correct Answer: C. Positive direct Coombs test

Question 103

Clinical Scenario:

A 48-year-old man presents to his primary care physician with diarrhea and weight loss. He states he has had diarrhea for the past several months that has been worsening steadily. The patient recently went on a camping trip and drank unfiltered stream water. Otherwise, the patient endorses a warm and flushed feeling in his face that occurs sporadically. His temperature is 97.2°F (36.2°C), blood pressure is 137/68 mmHg, pulse is 110/min, respirations are 14/min, and oxygen saturation is 98% on room air. Physical exam is notable for a murmur heard best over the left lower sternal border and bilateral wheezing on pulmonary exam. Which of the following is the best initial step in management?

Answer Choices:

A. Echocardiography B. Plasma free metanephrine levels C. Pulmonary function tests D. Urinary 5-hydroxyindoleacetic acid level

Correct Answer: D. Urinary 5-hydroxyindoleacetic acid level

Question 104

Clinical Scenario:

A 71-year-old man with type 2 diabetes mellitus comes to the physician because of a 9-month history of pain and stiffness in the right knee. He reports that the stiffness lasts approximately 10 minutes after waking up and that the pain is worse in the evening. There is no history of trauma. He is 175 cm (5 ft 9 in) tall and weighs 102 kg (225 lb); BMI is 33 kg/m². Examination of the right knee shows tenderness in the anteromedial joint line and crepitus during knee movement. Laboratory studies show an erythrocyte sedimentation rate of 15 mm/h and a serum uric acid concentration of 6.9 mg/dL. Which of the following is the most likely finding on imaging of the right knee?

Answer Choices:

A. Osteophytes and narrowing of the joint-space B. Marginal bony erosions and opacification of periarticular soft tissue C. Loculated epiphyseal cyst with thinning of the overlying cortex D. Bony ankylosis and bone proliferation at the entheses

Correct Answer: A. Osteophytes and narrowing of the joint-space

Question 105

Clinical Scenario:

A 23-year-old man presents to an outpatient psychiatrist complaining of anxiety and a persistent feeling that “something terrible will happen to my family.” He describes 1 year of vague, disturbing thoughts about his family members contracting a “horrible disease” or dying in an accident. He believes that he can prevent these outcomes by washing his hands of “the contaminants” any time that he touches something and by performing praying and counting rituals each time that he has unwanted, disturbing thoughts. The thoughts and rituals have become more frequent recently, making it impossible for him to work, and he expresses feeling deeply embarrassed by them. Which of the following is the most effective treatment for this patient's disorder?

Answer Choices:

A. Cognitive behavioral therapy and clonazepam B. Cognitive behavioral therapy and fluoxetine C. Psychodynamic psychotherapy and citalopram D. Psychodynamic psychotherapy and aripiprazole

Correct Answer: B. Cognitive behavioral therapy and fluoxetine

Question 106

Clinical Scenario:

A 32-year-old woman presents complaining of nightmares. She reports that these “nightmares” happen when she is asleep and also sometimes when she is awake, but she cannot come up with another description for them. The episodes have been happening for at least 3 weeks now. She states that when it happens she feels “outside of her body,” like she’s “watching myself.” She also reports some chronic fatigue. The patient denies headaches, vision changes, dizziness, or loss in sensation or motor function. She has no notable medical or surgical history. She takes a multivitamin every day. She smokes 1 pack of cigarettes a day but denies alcohol or illicit drug use. The patient appears slightly anxious but is in no acute distress. A physical and neurological exam is normal. She denies suicidal or homicidal ideation. Which of the following is the most likely diagnosis for the patient’s symptoms?

Answer Choices:

A. Brief psychotic disorder B. Delusional disorder C. Depersonalization disorder D. Dissociative fugue disorder

Correct Answer: C. Depersonalization disorder

Question 107

Clinical Scenario:

A 26-year-old woman comes to the physician because of severe pain in her right wrist one day after falling onto her hands and knees while rollerskating. Physical examination shows abrasions over the knees and bruising over the volar aspect of the right wrist. There is swelling and tenderness on palpation of the volar wrist joint, as

well as restricted range of motion due to pain. An x-ray of the hand shows volar dislocation of the lunate bone. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Paresthesia over the volar aspect of the first 3 fingers on wrist flexion B. Anesthesia over the dorsal aspect of the first 3 fingers C. Tenderness to palpation of the anatomic snuffbox D. Pale skin color on the volar surface when pressure is applied to the radial artery

Correct Answer: A. Paresthesia over the volar aspect of the first 3 fingers on wrist flexion

Question 108

Clinical Scenario:

A healthy 22-year-old male participates in a research study you are leading to compare the properties of skeletal and cardiac muscle. You conduct a 3-phased experiment with the participant. In the first phase, you get him to lift up a 2.3 kg (5 lb) weight off a table with his left hand. In the second phase, you get him to do 20 burpees, taking his heart rate to 150/min. In the third phase, you electrically stimulate his gastrocnemius with a frequency of 50 Hz. You are interested in the tension and electrical activity of specific muscles as follows: Biceps in phase 1, cardiac muscle in phase 2, and gastrocnemius in phase 3. What would you expect to be happening in the phases and the respective muscles of interest?

Answer Choices:

A. Recruitment of small motor units at the start of experiments 1 and 2 B. Recruitment of large motor units followed by small motor units in experiment 1 C. Fused tetanic contraction at the end of all three experiments D. Increase of tension in all phases

Correct Answer: D. Increase of tension in all phases

Question 109

Clinical Scenario:

A 55-year-old woman who works as a chef is brought to the hospital for evaluation of burns sustained in a kitchen accident. Physical examination reveals 3rd-degree burns over the anterior surface of the right thigh and the lower limbs, which involve approx. 11% of the total body surface area (TBSA). The skin in the burned areas is thick and painless to touch, and the dorsalis pedis pulses are palpable but weak. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Early excision and split-thickness skin grafting B. Delayed excision and skin grafting C. Topical antibiotic application of mafenide acetate D. Fluid resuscitation with Ringer's lactate solution per the Parkland formula

Correct Answer: A. Early excision and split-thickness skin grafting

Question 110

Clinical Scenario:

A 56-year-old man comes to the physician because of chest pain and shortness of breath for 3 days. The pain is present at rest and worsens with deep inspiration. His temperature is 37.2°C (99°F), pulse is 102/min, respirations are 23/min, and blood pressure is 135/88 mm Hg. Examination shows decreased breath sounds at the left lower lobe. Laboratory studies show: Hematocrit 42% Leukocyte count 6,500/ μ L Serum Fasting glucose 90 mg/dL Lactate dehydrogenase 75 U/L Total protein 7.2 g/dL An x-ray of the chest shows a small left-sided pleural effusion but no other abnormalities. A diagnostic thoracentesis is performed and 100 mL of bloody fluid are aspirated from the left pleural space. Pleural fluid analysis shows a lactate dehydrogenase of 65 U/L and a total protein of 5.1 g/dL. Pleural fluid cytology shows normal cell morphology. Further evaluation of this patient is most likely to show a history of which of the following?"

Answer Choices:

A. Oropharyngeal dysphagia B. Infliximab use C. Prolonged immobilization D. Congestive heart failure

Correct Answer: C. Prolonged immobilization

Question 111

Clinical Scenario:

A 70-year-old man comes to the physician for evaluation of worsening nocturia, fatigue, and shortness of breath on exertion. While he used to be able to walk for 15 minutes at a time, he now has to pause every 5 minutes. Recently, he has started using two pillows to avoid waking up short of breath at night. He has a history of hypertension treated with daily amlodipine and prazosin; he has difficulty adhering to his medication regimen. His pulse is 75/min, and blood pressure is 150/90 mm Hg. Physical examination shows a laterally displaced apical heartbeat and 2+ bilateral pitting edema of the lower legs. Auscultation shows an S4 gallop and fine bibasilar rales. Further evaluation is most likely to show which of the following pathophysiologic changes in this patient?

Answer Choices:

A. Constriction of efferent renal arterioles B. Decrease in total peripheral vascular resistance C. Reduction of alveolar surface tension D. Retention of potassium

Correct Answer: A. Constriction of efferent renal arterioles

Question 112

Clinical Scenario:

A 55-year-old man presents with severe fatigue and fever. His past medical history is significant for a recent history of mononucleosis from which he fully recovered 8 weeks ago. On physical examination, the patient seems pale. A chest radiograph shows multiple enlarged mediastinal lymph nodes. A biopsy of one of the

enlarged mediastinal lymph nodes is performed and shows the presence of multinucleated cells with an 'owl-eye' appearance in a hypocellular background. This patient's most likely condition is very aggressive and associated with a very poor prognosis. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Lymphocyte-depleted lymphoma B. Diffuse large B cell lymphoma C. Follicular lymphoma D. Extranodal marginal zone lymphoma

Correct Answer: A. Lymphocyte-depleted lymphoma

Question 113

Clinical Scenario:

The parents of a 4-year-old present to the pediatrician because they are concerned about the poor growth and odd behavior of their son. Their son has been at the 10th percentile for growth since birth and they have noticed that his skin seems to have a bluish hue to it whenever he cries or is agitated. Recently, they have noticed that when he squats it seems to relieve these symptoms. What was the embryologic cause of this patient's current symptoms?

Answer Choices:

A. Failure of the aorticopulmonary septum to spiral B. Failure of the ductus arteriosus to obliterate C. Partial aorticopulmonary septum development D. Anterosuperior displacement of aorticopulmonary septum

Correct Answer: D. Anterosuperior displacement of aorticopulmonary septum

Question 114

Clinical Scenario:

A 19-year-old man is brought to the emergency department by the resident assistant of his dormitory for strange behavior. He was found locked out of his room, where the patient admitted to attending a fraternity party before becoming paranoid that the resident assistant would report him to the police. The patient appears anxious. His pulse is 105/min, and blood pressure is 142/85 mm Hg. Examination shows dry mucous membranes and bilateral conjunctival injection. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Tactile hallucinations B. Pupillary constriction C. Synesthesia D. Impaired reaction time

Correct Answer: D. Impaired reaction time

Question 115

Clinical Scenario:

A 34-year-old man comes to the physician because of blurry vision and fatigue for 2 months. During this period, he has also had occasional bleeding from his gums after brushing his teeth. One month ago, he was diagnosed with deep vein thrombosis after returning from an overseas business meeting. His pulse is 118/min, respirations are 19/min, and blood pressure is 149/91 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 97%. Examination shows bluish discoloration of the lips. The tip of the spleen is palpable 1 cm below the left costal margin. Sensory examination of the hands shows paresthesia. Hemoglobin concentration is 18 g/dL, hematocrit is 65%, leukocytes are 15,000/ μ L, and platelets are 470,000/ μ L. His serum erythropoietin concentration is decreased. Activation of which of the following is the most likely underlying cause of this patient's condition?

Answer Choices:

A. Cytokine receptor B. Antiapoptotic molecule C. Nonreceptor tyrosine kinase D. Serine/threonine kinase

Correct Answer: C. Nonreceptor tyrosine kinase

Question 116

Clinical Scenario:

A 68-year-old woman comes to the physician because of lower back pain that began suddenly 2 weeks ago after getting up from her chair. She has hypertension, chronic bilateral knee pain, and a history of breast cancer 15 years ago that was treated with lumpectomy. Her mother has rheumatoid arthritis. Medications include hydrochlorothiazide and acetaminophen. She appears well. Her vital signs are within normal limits. Physical examination shows tenderness to palpation of the lower spine. Both knees are enlarged and swollen. Neurologic examination shows sensorineural hearing impairment of the left ear. Her gamma-glutamyl transferase (GGT) is 30 U/L (N: 0–30 U/L), alkaline phosphatase (ALP) is 310 U/L, and serum calcium is 10.2 mg/dL. A spinal x-ray shows a fracture in the L4 vertebra. Which of the following is the most likely diagnosis?

Answer Choices:

A. Type 1 osteopetrosis B. Primary biliary cholangitis C. Paget disease of bone D. Bone metastases

Correct Answer: C. Paget disease of bone

Question 117

Clinical Scenario:

A 21-year-old female presents to the family physician with 3 weeks of headaches, sweating, and palpitations. Her BP was 160/125 mmHg, and a 24-hour urine test revealed elevated vanillylmandelic acid (VMA) and normetanephrine. Past medical history is notable for bilateral retinal hemangioblastomas, and family history is significant for three generations (patient, mother, and maternal grandfather) with similar symptoms. Genetic analysis revealed a mutation of a gene on chromosome 3p. Which of the following is the patient at risk of developing?

Answer Choices:

A. Clear cell renal cell carcinoma B. Retinoblastoma C. Osteosarcoma D. Breast cancer

Correct Answer: A. Clear cell renal cell carcinoma

Question 118

Clinical Scenario:

A 23-year-old woman comes to the physician because of vaginal discharge for 3 days. She has been sexually active with three male partners over the past year and uses condoms inconsistently. Her only medication is an oral contraceptive. Physical exam shows thin grayish-white vaginal discharge. There is no erythema of the vaginal mucosa. The pH of the discharge is 5.9. Adding potassium hydroxide (KOH) to a mount containing vaginal discharge produces a fishy odor. Further evaluation of this patient's vaginal discharge is most likely to show which of the following findings?

Answer Choices:

A. Gram-variable rod B. Spiral-shaped bacteria C. Gram-negative diplococci D. Flagellated protozoa

Correct Answer: A. Gram-variable rod

Question 119

Clinical Scenario:

A 3255-g (7-lb) female newborn is delivered at term. Pregnancy and delivery were uncomplicated. On the day of her birth, she is given a routine childhood vaccine that contains a noninfectious glycoprotein. This vaccine will most likely help prevent infection by which of the following pathogens?

Answer Choices:

A. Poliovirus B. Bordetella pertussis C. Rotavirus D. Hepatitis D virus "

Correct Answer: D. Hepatitis D virus "

Question 120

Clinical Scenario:

A 31-year-old woman comes to the physician because she thinks that her "right wrist is broken." She says that she has severe pain and that "the bone is sticking out." She has not had any trauma to the wrist. Her medical records indicate that she was diagnosed with schizophrenia 2 years ago and treated with olanzapine; she has not filled any prescriptions over the past 4 months. Three weeks ago, she stopped going to work because she "did not feel like getting up" in the morning. Vital signs are within normal limits. Physical examination of the right wrist shows no visible injury; there is no warmth, swelling, or erythema. Range of motion is limited by pain. On mental status examination, she has a flat affect. Her speech is pressured and she frequently changes the topic. She has short- and long-term memory deficits. Attention and concentration are poor. There is no evidence

of suicidal ideation. Urine toxicology screening is negative. An x-ray of the wrist shows no abnormalities. Which of the following is the most appropriate response to this patient's concerns?

Answer Choices:

A. "I cannot see any injury of your wrist and the physical exam as well as the x-ray don't show any injury. I imagine that feeling as if your wrist was broken may be very uncomfortable. Can you tell me more about what it feels like?" B. ""I understand your concerns; however, your symptoms seem to be psychological in nature. I would be happy to refer you to a mental health professional."" C. "It seems as though you are having a schizophrenia relapse. If you don't follow my recommendations and take your medications, you will most likely have further and possibly more severe episodes." D. ""You are clearly distressed. However, your tests do not suggest a physical problem that can be addressed with medications or surgery. I suggest that we meet and evaluate your symptoms on a regular basis.""

Correct Answer: A. "I cannot see any injury of your wrist and the physical exam as well as the x-ray don't show any injury. I imagine that feeling as if your wrist was broken may be very uncomfortable. Can you tell me more about what it feels like?"

Question 121

Clinical Scenario:

A newborn girl is rushed to the neonatal ICU after an emergency cesarean section due to unstable vital signs after delivery. The pregnancy was complicated due to oligohydramnios and pulmonary hypoplasia. Gestation was at 38 weeks. APGAR scores were 6 and 8 at 1 and 5 minutes respectively. The newborn's temperature is 37.0°C (98.6°F), the blood pressure is 60/40 mm Hg, the respiratory rate is 45/min, and the pulse is 140/min. Physical examination reveals irregularly contoured bilateral abdominal masses. Abdominal ultrasound reveals markedly enlarged echogenic kidneys (5 cm in the vertical dimension) with multiple cysts in the cortex and medulla. This patient is at highest risk of which of the following complications?

Answer Choices:

A. Subarachnoid hemorrhage B. Portal hypertension C. Recurrent UTI D. Self mutilation

Correct Answer: B. Portal hypertension

Question 122

Clinical Scenario:

A 4-year-old boy presents with a history of recurrent bacterial infections, including several episodes of pneumococcal sepsis. His family history is significant for 2 maternal uncles who died from similar symptoms. Laboratory tests reveal undetectable serum levels of all isotypes of immunoglobulins and reduced levels of B cells. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Bruton agammaglobulinemia B. Common variable immunodeficiency C. DiGeorge syndrome D. Hereditary angioedema

Correct Answer: A. Bruton agammaglobulinemia

Question 123

Clinical Scenario:

A 31-year-old physician notices that her senior colleague has been arriving late for work for the past 2 weeks. The colleague recently lost his wife to cancer and has been taking care of his 4 young children. Following the death of his wife, the department chair offered him extended time off, but he declined. Resident physicians have noted and discussed some recent changes in this colleague, such as missed clinic appointments, 2 intra-operative errors, and the smell of alcohol on his breath on 3 different occasions. Which of the following is the most appropriate action by the physician regarding her colleague?

Answer Choices:

A. Advise resident physicians to report future misconduct to the department chair B. Alert the State Licensing Board C. Confront the colleague in private D. Inform the local Physician Health Program

Correct Answer: D. Inform the local Physician Health Program

Question 124

Clinical Scenario:

A 52-year-old woman presents to the emergency room complaining of chest pain. She reports a 4-hour history of dull substernal pain radiating to her jaw. Her history is notable for hypertension, diabetes mellitus, and alcohol abuse. She has a 30 pack-year smoking history and takes lisinopril and metformin but has an allergy to aspirin. Her temperature is 99.1°F (37.3°C), blood pressure is 150/90 mmHg, pulse is 120/min, and respirations are 22/min. Physical examination reveals a diaphoretic and distressed woman. An electrocardiogram reveals ST elevations in leads I, aVL, and V5-6. She is admitted with plans for immediate transport to the catheterization lab for stent placement. What is the mechanism of the next medication that should be given to this patient?

Answer Choices:

A. Thrombin inhibitor B. Vitamin K epoxide reductase inhibitor C. ADP receptor inhibitor D. Cyclooxygenase activator

Correct Answer: C. ADP receptor inhibitor

Question 125

Clinical Scenario:

A 56-year-old man with chronic kidney disease and type 2 diabetes mellitus is brought to the emergency department by his neighbor because of impaired consciousness and difficulty speaking for 1 hour. A diagnosis of acute ischemic stroke is made. Over the next three days after admission, the patient's renal function slowly worsens and hemodialysis is considered. He is not alert and cannot communicate. The neighbor, who has been a close friend for many years, says that the patient has always emphasized he would refuse dialysis or any other life-prolonging measures. He also reports that the patient has no family besides his father, who he has not seen for many years. His wife died 2 years ago. Which of the following is the most appropriate action by the physician?

Answer Choices:

A. Avoid dialysis in line with the patient's wishes B. Start dialysis when required C. Try to contact the father for consent D. Consult ethics committee

Correct Answer: C. Try to contact the father for consent

Question 126

Clinical Scenario:

A 7-year-old boy is brought to the physician for the evaluation of sore throat for the past 2 days. During this period, he has had intermittent nausea and has vomited once. The patient has no cough, hoarseness, or rhinorrhea. He had similar symptoms at the age of 5 years that resolved spontaneously. He is otherwise healthy. His temperature is 37.9°C (100.2°F), pulse is 85/min, and blood pressure is 108/70 mm Hg. Head and neck examination shows an erythematous pharynx with grayish exudates overlying the palatine tonsils. There is no lymphadenopathy. Rapid antigen detection test for group A streptococci is negative. Which of the following is most appropriate next step in the management of this patient?

Answer Choices:

A. Measurement of antistreptolysin O titer B. Measurement of antiviral capsid antigen IgM antibody C. Penicillin V therapy D. Obtain throat culture "

Correct Answer: D. Obtain throat culture "

Question 127

Clinical Scenario:

A 67-year-old female presents to the emergency room with dry cough and malaise. She has no other complaints. She has a past medical history of a meningioma status post resection complicated by hemiplegia and has been managed with dexamethasone for several months. Her vital signs are T 100.4 F (38 C), O2 93% on room air, RR 20, BP 115/75 mmHg. Physical examination is notable for crackles bilaterally. A chest radiograph is obtained (Image A). The patient is admitted and initially treated guideline-compliant antibiotics for community-acquired pneumonia. Unfortunately, her respiratory function deteriorates. An arterial blood gas is drawn. On room air at sea level, PaO2 is 71 mmHg and PaCO2 is 34 mmHg. Induced sputum samples reveal organisms on methenamine silver stain. What is the best treatment strategy for this patient?

Answer Choices:

- A. Trimethoprim-sulfamethoxazole B. Trimethoprim-sulfamethoxazole + steroids C. Piperacillin-tazobactam
D. Piperacillin-tazobactam + steroids

Correct Answer: B. Trimethoprim-sulfamethoxazole + steroids

Question 128**Clinical Scenario:**

A 45-year-old female with a history of gastroesophageal reflux disease presents to her family physician with symptoms of epigastric pain right after a meal. The physician performs a urea breath test which is positive and the patient is started on appropriate medical therapy. Three days later at a restaurant, she experienced severe flushing, tachycardia, hypotension, and vomiting after her first glass of wine. Which of the following is the mechanism of action of the medication causing this side effect?

Answer Choices:

- A. Blocks protein synthesis by binding to the 50S ribosomal subunit inhibiting protein translocation B. Binds to the 30S ribosomal subunit preventing attachment of the aminoacyl-tRNA C. Forms toxic metabolites that damage bacterial DNA D. Inhibits the H⁺/K⁺ ATPase

Correct Answer: C. Forms toxic metabolites that damage bacterial DNA

Question 129**Clinical Scenario:**

A 43-year-old woman comes to the physician because of a 6-week history of hoarseness and difficulty swallowing. She also has a history of lower back pain treated with ibuprofen. She is 162 cm (5 ft 4 in) tall and weighs 77 kg (169 lb); BMI is 29 kg/m². Her pulse is 64/min and blood pressure is 130/86 mm Hg. Physical examination shows dry skin, swelling of the lower extremities, and a hard nontender anterior neck swelling that does not move with swallowing. Femoral and pedal pulses are decreased bilaterally. A biopsy of the neck swelling is most likely to show which of the following findings?

Answer Choices:

- A. Spindle cells and areas of focal necrosis B. Collagen deposition and decreased number of follicles C. Giant cells and noncaseating granulomas D. Lymphocytic infiltrate and germinal center formation

Correct Answer: B. Collagen deposition and decreased number of follicles

Question 130**Clinical Scenario:**

A 62-year-old woman is hospitalized after a recent viral illness complicated by congestive heart failure. She has a past medical history of obesity and hypertension controlled on lisinopril but was otherwise healthy until she developed fatigue and edema after a recent viral illness. In the hospital, she is started on furosemide to manage her fluid status. On day 5 of her admission, the patient's temperature is 100.0°F (37.8°C), blood pressure is 136/88 mmHg, pulse is 90/min, and respirations are 14/min. The patient continues to have normal heart sounds, but with crackles bilaterally on lung auscultation. Edema is 3+ up to the bilateral knees. On labs, her leukocyte count is now 13,000/mm³, up from 9,000/mm³ the day before. Differential shows that this includes 1,000 eosinophils/mm³. Creatinine is 1.7 mg/dL from 1.0 mg/dL the day before. Which of the following is most likely expected on urinary analysis?

Answer Choices:

A. Bacteria > 100 CFU/mL B. Crystals C. Leukocyte esterase positive D. Red blood cell casts

Correct Answer: C. Leukocyte esterase positive

Question 131

Clinical Scenario:

An 11-year-old boy is brought to the emergency department 30 minutes after he was found screaming and clutching his head. He has had nausea and occasional episodes of vomiting for 1 week, fever and left-sided headaches for 2 weeks, and increasing tooth pain over the past 3 weeks. He has no history of ear or sinus infections. He is in moderate distress. His temperature is 38.7°C (101.7°F), pulse is 170/min, respirations are 19/min, and blood pressure is 122/85 mmHg. He is confused and only oriented to person. The pupils react sluggishly to light. Fundoscopic examination shows papilledema bilaterally. Extraocular movements are normal. Flexion of the neck causes hip flexion. Which of the following is the most likely diagnosis?

Answer Choices:

A. Medulloblastoma B. HSV encephalitis C. Pyogenic brain abscess D. Cavernous sinus thrombosis

Correct Answer: C. Pyogenic brain abscess

Question 132

Clinical Scenario:

A 25-year-old woman is brought to the emergency department 12 hours after ingesting 30 tablets of an unknown drug in a suicide attempt. The tablets belonged to her father, who has a chronic heart condition. She has had nausea and vomiting. She also reports blurring and yellowing of her vision. Her temperature is 36.7°C (98°F), pulse is 51/min, and blood pressure is 108/71 mm Hg. Abdominal examination shows diffuse tenderness with no guarding or rebound. Bowel sounds are normal. An ECG shows prolonged PR-intervals and flattened T-waves. Further evaluation is most likely to show which of the following electrolyte abnormalities?

Answer Choices:

A. Increased serum Na⁺ B. Decreased serum K⁺ C. Decreased serum Na⁺ D. Increased serum K⁺

Correct Answer: D. Increased serum K⁺

Question 133

Clinical Scenario:

A 34-year-old woman comes to the physician because of a 6-week history of depressed mood, loss of interest, and difficulty sleeping. She also has had a 4.5-kg (10-lb) weight loss during this period. She has not been as productive as before at work due to difficulty concentrating. There is no evidence of suicidal ideation. Laboratory studies including thyroid-stimulating hormone are within the reference range. The physician prescribes treatment with escitalopram. This drug targets a neurotransmitter that is produced in which of the following brain structures?

Answer Choices:

A. Locus coeruleus B. Basal nucleus of Meynert C. Nucleus accumbens D. Raphe nucleus

Correct Answer: D. Raphe nucleus

Question 134

Clinical Scenario:

A 3-year-old girl presents to the emergency department with skin desquamation over her hips and buttocks and right arm; she also has conjunctivitis and fever. The patient was previously seen by her pediatrician for symptoms of impetigo around the nasal folds, and she was treated with topical fusidic acid. She was born at 39 weeks' gestation via spontaneous vaginal delivery, is up to date on all vaccines, and is meeting all developmental milestones. Medical history and family history are unremarkable. She is admitted to the hospital and started on IV antibiotics. Today, her blood pressure is 100/60 mm Hg, heart rate is 100 beats per minute, respiratory rate is 22 breaths per minute, and temperature is 39.4°C (102.9°F). The total area of desquamation exceeds 20%, sparing the mucous membranes. She is transferred to the pediatric intensive care unit. What is the most likely cause of the disease?

Answer Choices:

A. Herpes simplex virus infection B. Staphylococcus aureus infection C. Bullous pemphigoid D. Psoriasis

Correct Answer: B. Staphylococcus aureus infection

Question 135

Clinical Scenario:

An 8-year-old boy is brought to the physician for evaluation of developmental delay and recurrent tonic-clonic seizures. There is no family history of seizures or other serious illness. Current medications include risperidone for hyperactivity. He is at the 17th percentile for head circumference. Examination shows protrusion of the

mandible, strabismus, and a laughing facial expression. His gait is unsteady. He has a vocabulary of about 200 words and cannot speak in full sentences. Karyotype analysis shows a 46, XY karyotype without chromosomal deletions. Which of the following genetic mechanisms best explains this patient's findings?

Answer Choices:

A. Chromosome 22q11 microdeletion B. De novo mutation of MECP2 on the X chromosome C. Uniparental disomy of chromosome 15 D. Trinucleotide repeat in FMR1 gene

Correct Answer: C. Uniparental disomy of chromosome 15

Question 136

Clinical Scenario:

A 55-year-old man, who was recently diagnosed with tuberculosis, presents to his primary care provider as part of his routine follow-up visit every month. He is currently in the initial phase of anti-tubercular therapy. His personal and medical histories are relevant for multiple trips to Southeast Asia as part of volunteer activities and diabetes of 5 years duration, respectively. A physical examination is unremarkable except for a visual abnormality on a color chart; he is unable to differentiate red from green. The physician suspects the visual irregularity as a sign of toxicity due to one of the drugs in the treatment regimen. Which of the following is the mechanism by which this medication acts in the treatment of *Mycobacterium tuberculosis*?

Answer Choices:

A. Inhibition of mycolic acid synthesis B. Induction of free radical metabolites C. Inhibition of protein synthesis by binding to the 30S ribosomal subunit D. Inhibition of arabinosyltransferase

Correct Answer: D. Inhibition of arabinosyltransferase

Question 137

Clinical Scenario:

A 75-year-old woman presents complaining of severe shortness of breath and peripheral edema. Her family reports that she has gained a significant amount of weight within the past week. Despite considerable efforts in the emergency department and ICU, she dies from sudden cardiac death overnight. The family requests an autopsy to determine her cause of death. Amongst other studies, a biopsy of her liver is shown. What was the most likely cause of the liver changes shown?

Answer Choices:

A. Budd-Chiari syndrome B. Congestive heart failure C. Hepatic metastasis D. Amebic liver abscess

Correct Answer: B. Congestive heart failure

Question 138

Clinical Scenario:

A 5-year-old boy is brought to the physician by his parents because of a 4-day history of arthralgias, abdominal pain, and lesions on his arms and legs. Ten days ago, he had an upper respiratory tract infection. A photograph of one of his legs is shown. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Genital ulcers B. Hematuria C. Tick bite D. Thrombocytopenia

Correct Answer: B. Hematuria

Question 139

Clinical Scenario:

A 72-year-old man comes to the physician for a routine physical examination. He says that he has felt well except for occasional headaches. He has no history of major medical illness. His temperature is 37°C (98.6°F), pulse is 80/min, and blood pressure is 155/90 mm Hg. An ultrasound of the kidneys shows a normal right kidney and a left kidney that is 2 cm smaller in length. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Elevated urine metanephrines B. Abdominal bruit C. Polycythemia D. Hematuria

Correct Answer: B. Abdominal bruit

Question 140

Clinical Scenario:

A newborn presents with central cyanosis, nasal flaring, and subcostal retractions following a scheduled cesarean delivery. He was born to a healthy 29-year-old G-1-P-1 mother due to cervical incompetence at 34 weeks gestation; the pregnancy was otherwise uneventful. Appearance, Pulse, Grimace, Activity, and Respiration (APGAR) scores were 6 and 8 at 1 and 5 minutes, respectively and his birth weight was 3,200 g. The umbilical cord had 3 vessels and the placenta was tan-red with all cotyledons intact. Fetal membranes were tan-white and semi-translucent. Currently, the vital signs include: temperature 36.9°C (98.4°F), blood pressure 70/40 mm Hg, pulse 190/min, and respiratory rate 68/min. On auscultation, breath sounds are decreased. Diffuse ground-glass opacifications are identified on chest X-ray. Which of the factors listed below most likely contributed to this infant's current condition?

Answer Choices:

A. Gestational diabetes B. Alcohol abuse C. Down syndrome D. Lithium

Correct Answer: A. Gestational diabetes

Question 141

Clinical Scenario:

A 60-year-old man comes to the emergency department because of a 3-day history of fever and shortness of breath. He has a history of COPD treated with inhaled albuterol. His temperature is 39.0°C (102.2°F), pulse is 95/min, respirations are 20/min, and blood pressure is 130/80 mm Hg. Cardiopulmonary examination shows decreased breath sounds and poor air movement over the left lung. A lateral decubitus x-ray of the chest shows a pleural effusion height of 2 cm. Thoracentesis is performed and pleural fluid analysis shows a protein concentration of 4.0 g/dL and LDH of 80 U/L. Which of the following is the most likely underlying cause of this patient's effusion?

Answer Choices:

A. Impaired lymphatic flow B. Increased pulmonary capillary permeability C. Decreased intrapleural pressure D. Decreased plasma oncotic pressure

Correct Answer: B. Increased pulmonary capillary permeability

Question 142

Clinical Scenario:

A 27-year-old woman, gravida 2, para 1, at 26 weeks' gestation comes to the emergency department because of vaginal bleeding and epistaxis for the past 2 days. She missed her last prenatal visit 2 weeks ago. Physical examination shows blood in the posterior pharynx and a uterus consistent in size with 23 weeks' gestation. Her hemoglobin concentration is 7.2 g/dL. Ultrasonography shows an intrauterine pregnancy with a small retroplacental hematoma and absent fetal cardiac activity. Further evaluation is most likely to show which of the following findings?

Answer Choices:

A. Increased antithrombin concentration B. Decreased prothrombin time C. Increased factor V concentration D. Decreased fibrinogen concentration

Correct Answer: D. Decreased fibrinogen concentration

Question 143

Clinical Scenario:

A 57-year-old man presents with 2 days of severe, generalized, abdominal pain that is worse after meals. He is also nauseated and reports occasional diarrhea mixed with blood. Apart from essential hypertension, his medical history is unremarkable. His vital signs include a temperature of 36.9°C (98.4°F), blood pressure of 145/92 mm Hg, and an irregularly irregular pulse of 105/min. Physical examination is only notable for mild periumbilical tenderness. Which of the following is the most likely diagnosis?

Answer Choices:

A. Acute pancreatitis B. Crohn's disease C. Acute mesenteric ischemia D. Diverticular disease

Correct Answer: C. Acute mesenteric ischemia

Question 144**Clinical Scenario:**

A 60-year-old woman presents to the emergency room with chest pain that started 20 minutes ago while watching television at home. The pain is substernal and squeezing in nature. She rates the pain as 6/10 and admits to having similar pain in the past with exertion. Her past medical history is significant for diabetes mellitus that is controlled with metformin. The physical examination is unremarkable. An electrocardiogram (ECG) shows ST-segment depression in the lateral leads. She is started on aspirin, nitroglycerin, metoprolol, unfractionated heparin, and insulin. She is asked not to take metformin while at the hospital. Three sets of cardiac enzymes are negative. Lab results are given below: Serum glucose 88 mg/dL Sodium 142 mEq/L Potassium 3.9 mEq/L Chloride 101 mEq/L Serum creatinine 1.2 mg/dL Blood urea nitrogen 22 mg/dL Cholesterol, total 170 mg/dL HDL-cholesterol 40 mg/dL LDL-cholesterol 80 mg/dL Triglycerides 170 mg/dL Hematocrit 38% Hemoglobin 13 g/dL Leucocyte count 7,500/mm³ Platelet count 185,000 /mm³ Activated partial thromboplastin time (aPTT) 30 seconds Prothrombin time (PT) 12 seconds Urinalysis Glucose negative Ketones negative Leucocytes negative Nitrites negative Red blood cells (RBC) negative Casts negative An echocardiogram reveals left ventricular wall motion abnormalities. With the pain subsiding, she was admitted and the medications were continued. A coronary angiography is planned in 4 days. In addition to regular blood glucose testing, which of the following should be closely monitored in this patient?

Answer Choices:

A. Prothrombin time and platelet count B. Prothrombin time alone C. aPTT and platelet count D. Platelet count alone

Correct Answer: C. aPTT and platelet count

Question 145**Clinical Scenario:**

A 66-year-old man comes to the physician because of yellowish discoloration of his eyes and skin, abdominal discomfort, and generalized fatigue for the past 2 weeks. He has had dark urine and pale stools during this period. He has had a 10-kg (22-lb) weight loss since his last visit 6 months ago. He has hypertension. He has smoked one pack of cigarettes daily for 34 years. He drinks three to four beers over the weekends. His only medication is amlodipine. His temperature is 37.3°C (99.1°F), pulse is 89/min, respirations are 14/min, and blood pressure is 114/74 mm Hg. Examination shows jaundice of the sclera and skin and excoriation marks on his trunk and extremities. The lungs are clear to auscultation. The abdomen is soft and nontender. The remainder of the examination shows no abnormalities. Laboratory studies show: Hemoglobin 12 g/dL Leukocyte count 5,000/mm³ Platelet count 400,000/mm³ Serum Urea nitrogen 28 mg/dL Creatinine 1.2 mg/dL

Bilirubin Total 7.0 mg/dL Direct 5.5 mg/dL Alkaline phosphatase 615 U/L Aspartate aminotransferase (AST, GOT) 170 U/L Alanine aminotransferase (ALT, GPT) 310 U/L γ -Glutamyltransferase (GGT) 592 U/L (N = 5–50 U/L) An ultrasound shows extrahepatic biliary dilation. A CT scan of the abdomen shows a 2.5-cm (1-in) mass in the head of the pancreas with no abdominal lymphadenopathy. The patient undergoes biliary stenting. Which of the following is the most appropriate next step in the management of this patient?"

Answer Choices:

A. Stereotactic radiation therapy B. Pancreaticoduodenectomy C. Gemcitabine and 5-fluorouracil therapy D. Central pancreatectomy

Correct Answer: B. Pancreaticoduodenectomy

Question 146

Clinical Scenario:

A 71-year-old male presents to the emergency department after having a generalized tonic-clonic seizure. His son reports that he does not have a history of seizures but has had increasing confusion and weakness over the last several weeks. An electrolyte panel reveals a sodium level of 120 mEq/L and a serum osmolality of 248 mOsm/kg. His urine is found to have a high urine osmolality. His temperature is 37° C (98.6° F), respirations are 15/min, pulse is 67/min, and blood pressure is 122/88 mm Hg. On examination he is disoriented, his pupils are round and reactive to light and accommodation and his mucous membranes are moist. His heart has a regular rhythm without murmurs, his lungs are clear to auscultation bilaterally, the abdomen is soft, and his extremities have no edema but his muscular strength is 3/5 bilaterally. There is hyporeflexia of all four extremities. What is the most likely cause of his symptoms?

Answer Choices:

A. Syndrome of Inappropriate Antidiuretic Hormone (SIADH) B. Sheehan's syndrome C. Lithium use D. Diabetic ketoacidosis

Correct Answer: A. Syndrome of Inappropriate Antidiuretic Hormone (SIADH)

Question 147

Clinical Scenario:

A previously healthy, 16-year-old boy is brought to the emergency department with persistent bleeding from his gums after an elective removal of an impacted tooth. Multiple gauze packs were applied with minimal effect. He has a history of easy bruising. His family history is unremarkable except for a maternal uncle who had a history of easy bruising and joint swelling. Laboratory studies show: Hematocrit 36% Platelet count 170,000/mm³ Prothrombin time 13 sec Partial thromboplastin time 65 sec Bleeding time 5 min Peripheral blood smear shows normal-sized platelets. Which of the following is the most likely diagnosis?"

Answer Choices:

A. Von Willebrand disease B. Glanzmann thrombasthenia C. Hemophilia D. Bernard-Soulier syndrome

Correct Answer: C. Hemophilia

Question 148

Clinical Scenario:

A mother with HIV has given birth to a healthy boy 2 days ago. She takes her antiretroviral medication regularly and is compliant with the therapy. Before being discharged, her doctor explains that she cannot breastfeed the child since there is a risk of infection through breastfeeding and stresses that the child can benefit from formula feeding. The physician stresses the importance of not overheating the formula since Vitamin C may be inactivated by overheating. Which process could be impaired if the mother boiled the formula longer than needed?

Answer Choices:

A. Heme synthesis B. Collagen synthesis C. Protein catabolism D. Fatty acid metabolism

Correct Answer: B. Collagen synthesis

Question 149

Clinical Scenario:

A 56-year-old woman, gravida 3, para 3, comes to the physician because her left breast has become larger, hot, and itchy over the past 2 months. The patient felt a small lump in her left breast 1 year ago but did not seek medical attention at that time. She has hypertension and hyperlipidemia. Menarche was at the age of 11 years and menopause at the age of 46 years. Her mother died of breast cancer at the age of 45 years. The patient does not smoke or drink alcohol. Current medications include labetalol, simvastatin, and daily low-dose aspirin. She is 170 cm (5 ft 7 in) tall and weighs 78 kg (172 lb); BMI is 27 kg/m². Her temperature is 37.7°C (99.9°F), pulse is 78/min, and blood pressure is 138/88 mm Hg. Examination shows large dense breasts. There is widespread erythema and edematous skin plaques over a breast mass in the left breast. The left breast is tender to touch and left-sided axillary lymphadenopathy is noted. Which of the following is the most likely diagnosis?

Answer Choices:

A. Paget's disease of the breast B. Mastitis C. Breast abscess D. Inflammatory breast cancer

Correct Answer: D. Inflammatory breast cancer

Question 150

Clinical Scenario:

A 50-year-old man presents to the emergency department due to altered mental status. His symptoms began approximately two weeks prior to presentation where he complained of increasing fatigue, malaise, loss of appetite, and subjective fever. Vital signs are significant for a temperature of 102.0°F (38.9°C). On physical

examination, there is a holosystolic murmur in the tricuspid area, linear non-blanching reddish lesions under the nails, and needle tracks on both antecubital fossa. A transthoracic echocardiogram shows a vegetation on the tricuspid valve. Blood cultures return positive for *Staphylococcus aureus*. A lumbar puncture is prompted due to altered mental status in the setting of fever; however, there is no bacteria found on cerebral spinal fluid (CSF) culture. Which of the following cell structures prevents the penetration of the bacteria into the CSF from his blood?

Answer Choices:

A. Desmosomes B. Gap junctions C. Tight junctions D. Capillary fenestrations

Correct Answer: C. Tight junctions

Question 151

Clinical Scenario:

A 13-year-old boy is brought to the physician because of a 4-month history of worsening dizziness, nausea, and feeling clumsy. An MRI of the brain shows a well-demarcated, 4-cm cystic mass in the posterior fossa. The patient undergoes complete surgical resection of the mass. Pathologic examination of the surgical specimen shows parallel bundles of cells with eosinophilic, corkscrew-like processes. Which of the following is the most likely diagnosis?

Answer Choices:

A. Medulloblastoma B. Pilocytic astrocytoma C. Craniopharyngioma D. Pinealoma

Correct Answer: B. Pilocytic astrocytoma

Question 152

Clinical Scenario:

A 3-year-old African-American boy presents with a rapid onset of severe abdominal pain. He has a palpably enlarged mass in the left upper quadrant of his abdomen. Complete blood count is notable for a hemoglobin of 7.2 g/dL. Serum haptoglobin level returns normal. Serum unconjugated bilirubin is elevated. The corrected reticulocyte count is elevated. Which of the following is the most likely explanation for the findings above?

Answer Choices:

A. Aplastic crisis B. Renal infarction C. Intravascular hemolysis D. Extravascular hemolysis

Correct Answer: D. Extravascular hemolysis

Question 153

Clinical Scenario:

A 53-year-old man with a history of alcoholic liver cirrhosis was admitted to the hospital with ascites and general wasting. He has a history of 3-5 ounces of alcohol consumption per day for 20 years and 20-pack-year smoking history. Past medical history is significant for alcoholic cirrhosis of the liver, diagnosed 5 years ago. On physical examination, the abdomen is firm and distended. There is mild tenderness to palpation in the right upper quadrant with no rebound or guarding. Shifting dullness and a positive fluid wave is present. Prominent radiating umbilical varices are noted. Laboratory values are significant for the following: Total bilirubin 4.0 mg/dL Aspartate aminotransferase (AST) 40 U/L Alanine aminotransferase (ALT) 18 U/L Gamma-glutamyltransferase 735 U/L Platelet count 11,000/mm³ WBC 4,300/mm³ Serology for viral hepatitis B and C are negative. A Doppler ultrasound of the abdomen shows significant enlargement of the epigastric superficial veins and hepatofugal flow within the portal vein. There is a large volume of ascites present. Paracentesis is performed in which 10 liters of straw-colored fluid is removed. Which of the following sites of the portocaval anastomosis is most likely to rupture and bleed first in this patient?

Answer Choices:

A. Left branch of portal vein – inferior vena cava B. Esophageal branch of left gastric vein – esophageal branches of azygos vein C. Umbilical vein – superficial epigastric veins D. Superior and middle rectal vein – inferior rectal veins

Correct Answer: B. Esophageal branch of left gastric vein – esophageal branches of azygos vein

Question 154

Clinical Scenario:

A 52-year-old woman comes to the emergency department because of epigastric abdominal pain that started after her last meal and has become progressively worse over the past 6 hours. She has had intermittent pain similar to this before, but it has never lasted this long. Her temperature is 39°C (102.2°F). Examination shows a soft abdomen with normal bowel sounds. The patient has sudden inspiratory arrest during right upper quadrant palpation. Her alkaline phosphatase, total bilirubin, amylase, and aspartate aminotransferase levels are within the reference ranges. Abdominal imaging is most likely to show which of the following findings?

Answer Choices:

A. Dilated common bile duct with intrahepatic biliary dilatation B. Gallstone in the cystic duct C. Fistula formation between the gallbladder and bowel D. Decreased echogenicity of the liver

Correct Answer: B. Gallstone in the cystic duct

Question 155

Clinical Scenario:

A 66-year-old man presents to the emergency department with dyspnea. Two days ago, he hosted his grandchild's birthday party, and since has noticed general malaise, fever, and dry cough. He does not know if he feels more dyspneic while supine or standing but has noticed difficulty breathing even while watching television. He has a past medical history of congestive heart failure and hypertension, for which he takes

aspirin, metoprolol, furosemide, lisinopril, and spironolactone as prescribed. His blood pressure is 90/50 mmHg, pulse is 120/min, and respirations are 30/min. His radial pulse is barely palpable, and his wrists and ankles are cold and clammy. Physical exam reveals a S3 and S4 with a soft holosystolic murmur at the apex, decreased breath sounds up to the middle lung fields, jugular venous distention to the auricles, and 3+ pitting edema to the mid thighs. EKG shows ST depressions consistent with demand ischemia. Bedside echocardiogram shows global akinesis with an ejection fraction (EF) of 20%; previous reports show EF at 40%. A portable chest radiograph shows bilateral pulmonary edema. Metoprolol is held, dobutamine and furosemide drips are started, and BiPAP is started at 20/5 cm H₂O. After 15 minutes, the nurse reports that urine output is minimal and blood pressure is now 75/40 mmHg and pulse is 130/min. What is the best next step in management?

Answer Choices:

A. Resume home metoprolol B. Decrease furosemide rate C. Decrease dobutamine rate D. Decrease positive inspiratory pressure

Correct Answer: D. Decrease positive inspiratory pressure

Question 156

Clinical Scenario:

A 3-year-old boy is brought to the physician because of a 4-week history of generalized fatigue and malaise. He was born at term and has been healthy since. His mother has a history of recurrent anemia. He appears pale. His temperature is 37°C (98.6°F) and pulse is 97/min. Examination shows pale conjunctivae and jaundice. The abdomen is soft and nontender; the spleen is palpated 3–4 cm below the left costal margin. Laboratory studies show: Hemoglobin 9.3 g/dL Mean corpuscular volume 81.3 μ m³ Mean corpuscular hemoglobin concentration 39% Hb/cell Leukocyte count 7300/mm³ Platelet count 200,000/mm³ Red cell distribution width 19% (N = 13–15) Which of the following is most likely to confirm the diagnosis?"

Answer Choices:

A. Fluorescent spot test B. Eosin-5-maleimide binding test C. Indirect antiglobulin test D. Peripheral smear

Correct Answer: B. Eosin-5-maleimide binding test

Question 157

Clinical Scenario:

A 15-year-old girl is brought to the physician because of a 8-month history of fatigue, intermittent postprandial abdominal bloating and discomfort, foul-smelling, watery diarrhea, and a 7-kg (15-lb) weight loss. She developed a pruritic rash on her knees 3 days ago. Physical examination shows several tense, excoriated vesicles on the knees bilaterally. The abdomen is soft and nontender. Her hemoglobin concentration is 8.2 g/dL and mean corpuscular volume is 76 μ m³. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. IgA tissue transglutaminase antibodies B. Periodic acid-Schiff-positive macrophages C. Elevated serum amylase concentration D. Positive hydrogen breath test

Correct Answer: A. IgA tissue transglutaminase antibodies

Question 158**Clinical Scenario:**

A 65-year-old non-smoking woman with no symptoms comes to your clinic to establish care with a primary care provider. She hasn't seen a doctor in 12 years and states that she feels very healthy. You realize that guidelines by the national cancer organization suggest that she is due for some cancer screening tests, including a mammogram for breast cancer, a colonoscopy for colon cancer, and a pap smear for cervical cancer. These three screening tests are most likely to be considered which of the following?

Answer Choices:

A. Secondary prevention B. Tertiary prevention C. Quaternary prevention D. Cancer screening does not fit into these categories

Correct Answer: A. Secondary prevention

Question 159**Clinical Scenario:**

A 21-year-old man presents to the emergency room with abdominal pain and nausea for the past 5 hours. The pain is diffusely spread and of moderate intensity. The patient also says he has not felt like eating since yesterday. He has no past medical history and is not on any medications. He regularly drinks 2–4 beers per day but does not smoke or use illicit substances. Vitals show a pulse of 120/min, a respiratory rate of 26/min, a blood pressure of 110/60 mm Hg, and a temperature of 37.8°C (100.0°F). Examination reveals a soft, diffusely tender abdomen with no guarding. Bowel sounds are present. His mucous membranes are slightly dry and there is a fruity smell to his breath. Laboratory tests show: Laboratory test pH 7.31 Serum glucose (random) 450 mg/dL Serum electrolytes Sodium 149 mEq/L Potassium 5 mEq/L Chloride 99 mEq/L Bicarbonate 16 mEq/L Serum creatinine 1.0 mg/dL Blood urea nitrogen 15 mg/dL Urinalysis Proteins Negative Glucose Positive Ketones Positive Leucocytes Negative Nitrites Negative Red blood cells (RBC) Negative Casts Negative Which of the following explains this patient's presentation?

Answer Choices:

A. Blunt trauma to the abdomen B. Burn out of pancreatic beta cells C. Fecalith in the caecum D. Presence of gut contents in the abdominal cavity

Correct Answer: B. Burn out of pancreatic beta cells

Question 160

Clinical Scenario:

A 22-month-old girl is brought to the emergency department with a 24-hour history of fever, irritability, and poor feeding. The patient never experienced such an episode in the past. She met the normal developmental milestones, and her vaccination history is up-to-date. She takes no medications, currently. Her temperature is 38.9°C (102.0°F). An abdominal examination reveals general tenderness without organomegaly. The remainder of the physical examination shows no abnormalities. Laboratory studies show the following results: Urine Blood 1+ WBC 10–15/hpf Bacteria Many Nitrite Positive Urine culture from a midstream collection reveals 100,000 CFU/mL of Escherichia coli. Which of the following interventions is the most appropriate next step in evaluation?

Answer Choices:

A. Dimercaptosuccinic acid renal scan B. Intravenous pyelography C. Renal and bladder ultrasonography D. Voiding cystourethrography

Correct Answer: C. Renal and bladder ultrasonography

Question 161

Clinical Scenario:

An otherwise healthy 47-year-old woman comes to the physician for the evaluation of a 4-month history of worsening fatigue and constipation. She has also noticed that her cheeks appear fuller and her voice has become hoarse. Her temperature is 36.3°C (97.3°F) and pulse is 59/min. Examination of the neck shows a painless, mildly enlarged thyroid gland. Her skin is dry and cool and her nails appear brittle. Serum studies show antibodies against thyroid peroxidase. A biopsy of the thyroid gland is most likely to show which of the following?

Answer Choices:

A. Large, irregular nuclei, nuclear grooves, and Psammoma bodies B. Spindle cells, pleomorphic giant cells, and mitotic figures C. Multinucleated giant cells, macrophages, and degenerated follicular cells D. Lymphocytic infiltration, Hürthle cells, and germinal centers

Correct Answer: D. Lymphocytic infiltration, Hürthle cells, and germinal centers

Question 162

Clinical Scenario:

A 38-year-old female presents to her primary care physician with complaints of several episodes of palpitations accompanied by panic attacks over the last month. She also is concerned about many instances over the past few weeks where food has been getting stuck in her throat and she has had trouble swallowing. She denies any prior medical problems and reports a family history of cancer in her mother and maternal grandfather but

cannot recall any details regarding the type of cancer(s) or age of diagnosis. Her vital signs at today's visit are as follows: T 37.6 deg C, HR 106, BP 158/104, RR 16, SpO2 97%. Physical examination is significant for a nodule on the anterior portion of the neck that moves with swallowing, accompanied by mild lymphadenopathy. A preliminary work-up is initiated, which shows hypercalcemia, elevated baseline calcitonin, and an inappropriately elevated PTH level. Diagnostic imaging shows bilateral adrenal lesions on an MRI of the abdomen/pelvis. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Familial medullary thyroid cancer (FMTc) B. Li-Fraumeni syndrome C. Multiple endocrine neoplasia (MEN) IIa D. Multiple endocrine neoplasia (MEN) IIb

Correct Answer: C. Multiple endocrine neoplasia (MEN) IIa

Question 163

Clinical Scenario:

A 45-year-old man comes to the physician because of numbness and tingling in his fingers and toes for the past month. He also describes difficulty with balance while walking. Laboratory studies show a hemoglobin concentration of 9.5 g/dL. Serum homocysteine and methylmalonic acid levels are elevated. Peripheral blood smear shows hypersegmented neutrophils. Which of the following is most likely to have prevented this patient's condition?

Answer Choices:

A. Avoidance of canned foods B. Cyanocobalamin supplementation C. Pyridoxine supplementation D. Folic acid supplementation

Correct Answer: B. Cyanocobalamin supplementation

Question 164

Clinical Scenario:

A 19-year-old male college student is admitted to an inpatient psychiatric unit with a chief complaint of "thoughts about killing my girlfriend." The patient explains that throughout the day he becomes suddenly overwhelmed by thoughts about strangling his girlfriend and hears a voice saying "kill her." He recognizes the voice as his own, though it is very distressing to him. After having such thoughts, he feels anxious and guilty and feels compelled to tell his girlfriend about them in detail, which temporarily relieves his anxiety. He also worries about his girlfriend dying in various ways but believes that he can prevent all of this from happening and "keep her safe" by repeating prayers out loud several times in a row. The patient has no personal history of violence but has a family history of psychotic disorders. He has been on haloperidol and fluoxetine for his symptoms in the past but neither was helpful. In addition to psychotherapy, which of the following medications is the most appropriate treatment for this patient?

Answer Choices:

A. Alprazolam B. Amitriptyline C. Buspirone D. Clomipramine

Correct Answer: D. Clomipramine

Question 165

Clinical Scenario:

A 36-year-old woman is brought to the emergency room by her husband for "weird behavior" for the past several weeks. He reports that her right arm has been moving uncontrollably in a writhing movement and that she has been especially irritable. She has a history of depression, which was diagnosed 4 years ago and is currently being treated with sertraline. She denies any recent fever, trauma, infections, travel, weakness, or sensory changes. She was adopted so is unsure of her family history. Which of the following is the most likely explanation for her symptoms?

Answer Choices:

A. CAG triplet expansion on chromosome 4 B. Frontotemporal lobe degeneration C. GAA triplet expansion on chromosome 9 D. Presence of misfolded proteins in the brain

Correct Answer: A. CAG triplet expansion on chromosome 4

Question 166

Clinical Scenario:

The physician recommends that the patient receive an influenza vaccine. The patient becomes nervous and reports that he has never received an influenza vaccination because of an allergy to eggs. The allergy was diagnosed many years ago, after he developed hives upon eating scrambled eggs. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Administer inactivated influenza vaccine B. Administer influenza immunoglobulins C. End the examination without additional measures D. Prescribe oseltamivir for standby emergency treatment

Correct Answer: A. Administer inactivated influenza vaccine

Question 167

Clinical Scenario:

A 25-year-old male medical student presents to student health with a chief complaint of picking at his skin. He states that at times he has urges to pick his skin that he struggles to suppress. Typically, he will participate in the act during finals or when he has "too many assignments to do." The patient states that he knows that his behavior is not helping his situation and is causing him harm; however, he has trouble stopping. He will often ruminate over all his responsibilities which make his symptoms even worse. The patient has a past medical

history of surgical repair of his ACL two years ago. His current medications include melatonin. On physical exam you note a healthy young man with scars on his arms and face. His neurological exam is within normal limits. Which of the following is the best initial step in management?

Answer Choices:

A. Clomipramine B. Fluoxetine C. Dialectical behavioral therapy D. Interpersonal psychotherapy

Correct Answer: B. Fluoxetine

Question 168

Clinical Scenario:

A 43-year-old woman presents to her primary care provider for follow-up of her glucose levels. At her last visit 3 months ago, her fasting serum glucose was 128 mg/dl. At that time, she was instructed to follow a weight loss regimen consisting of diet and exercise. Her family history is notable for a myocardial infarction in her father and type II diabetes mellitus in her mother. She does not smoke and drinks 2-3 glasses of wine per week. Her temperature is 99°F (37.2°C), blood pressure is 131/78 mmHg, pulse is 80/min, and respirations are 17/min. Her BMI is 31 kg/m². On exam, she is well-appearing and appropriately interactive. Today, despite attempting to make the appropriate lifestyle changes, a repeat fasting serum glucose is 133 mg/dl. The patient is prescribed the first-line oral pharmacologic agent for her condition. Which of the following is the correct mechanism of action of this medication?

Answer Choices:

A. Activation of peroxisome proliferator-activating receptors B. Closure of potassium channels in pancreatic beta cells C. Inhibition of alpha-glucosidase in the intestinal brush border D. Inhibition of hepatic gluconeogenesis

Correct Answer: D. Inhibition of hepatic gluconeogenesis

Question 169

Clinical Scenario:

A 71-year-old man presents to the physician for a routine health-maintenance examination. He feels well; however, he is concerned about the need for prostate cancer screening. He has a 3-year history of benign prostatic hyperplasia. His symptoms of urinary hesitancy and terminal dribbling of urine are well controlled with tamsulosin and finasteride. He also had a percutaneous coronary angioplasty done 2 years ago following a diagnosis of unstable angina. His medication list also includes aspirin, atorvastatin, losartan, and nitroglycerin. His vital signs are within normal limits. He has never had a serum prostate-specific antigen (PSA) test or prostate ultrasonography. Which of the following is the most appropriate screening test for prostate cancer in this patient?

Answer Choices:

A. Prostate ultrasonography every year B. Serum PSA every year C. Serum PSA every 2-4 years D. No screening test is recommended

Correct Answer: D. No screening test is recommended

Question 170

Clinical Scenario:

A 61-year-old woman comes to the physician for a follow-up examination 1 week after undergoing right-sided radical mastectomy and axillary lymph node dissection for breast cancer. She says that she has been unable to comb her hair with her right hand since the surgery. Physical examination shows shoulder asymmetry. She is unable to abduct her right arm above 90 degrees. When she pushes against a wall, there is protrusion of the medial aspect of the right scapula. Injury to which of the following nerves is the most likely cause of this patient's condition?

Answer Choices:

A. Upper trunk of the brachial plexus B. Long thoracic nerve C. Suprascapular nerve D. Thoracodorsal nerve

Correct Answer: B. Long thoracic nerve

Question 171

Clinical Scenario:

A 36-year-old man comes to the physician because of a 2-week history of productive cough, weight loss, and intermittent fever. He recently returned from a 6-month medical deployment to Indonesia. He appears tired. Physical examination shows nontender, enlarged, palpable cervical lymph nodes. An x-ray of the chest shows right-sided hilar lymphadenopathy. A sputum smear shows acid-fast bacilli. A diagnosis of pulmonary tuberculosis is made from PCR testing of the sputum. The patient requests that the physician does not inform anyone of this diagnosis because he is worried about losing his job. Which of the following is the most appropriate initial action by the physician?

Answer Choices:

A. Inform the local public health department of the diagnosis B. Request the patient's permission to discuss the diagnosis with an infectious disease specialist C. Assure the patient that his diagnosis will remain confidential D. Confirm the diagnosis with a sputum culture

Correct Answer: A. Inform the local public health department of the diagnosis

Question 172

Clinical Scenario:

A 12-year-old boy is brought to the physician because of fever, malaise, and a painful, itchy rash on the right shoulder for 2 weeks. The patient's mother says the boy's condition has worsened over the past 4 days. He has a history of atopic dermatitis. He has lived with his mother at several public shelters since she separated from his physically abusive father 2 months ago. His immunizations are up-to-date. There is cervical lymphadenopathy. Laboratory studies show no abnormalities. A photograph of the rash is shown. Which of the following is the most likely diagnosis?

Answer Choices:

A. Bed bug bites B. Nonbullous impetigo C. Stevens-Johnson syndrome D. Eczema herpeticum

Correct Answer: D. Eczema herpeticum

Question 173

Clinical Scenario:

A 73-year-old man comes to the physician because of progressive fatigue and shortness of breath on exertion for 3 weeks. He has swelling of his legs. He has not had nausea or vomiting. His symptoms began shortly after he returned from a trip to Cambodia. He occasionally takes ibuprofen for chronic back pain. He has a history of arterial hypertension and osteoarthritis of both knees. He had an episode of pneumonia 4 months ago. His current medications include lisinopril and hydrochlorothiazide. He has no history of drinking or smoking. His temperature is 37°C (98.6°F), pulse is 101/min, and blood pressure is 135/76 mm Hg. Examination shows pitting edema of the upper and lower extremities. Laboratory studies show: Hemoglobin 14.1 g/dL Leukocyte count 6,800/mm³ Platelet count 216,000/mm³ Serum Urea nitrogen 26 mg/dL Creatinine 2.9 mg/dL Albumin 1.6 g/L Urine Blood negative Protein 4+ Glucose negative Renal biopsy with Congo red stain shows apple-green birefringence under polarized light. Further evaluation of this patient is most likely to show which of the following findings?"

Answer Choices:

A. Rouleaux formation on peripheral smear B. Elevated anti-citrullinated peptide antibodies C. Positive interferon-γ release assay D. Dilated bronchi on chest CT "

Correct Answer: A. Rouleaux formation on peripheral smear

Question 174

Clinical Scenario:

A 19-year-old woman presents for a sports physical. She says she feels healthy and has no concerns. Past medical history is significant for depression and seasonal allergies. Current medications are fluoxetine and oral estrogen/progesterone contraceptive pills. Family history is significant for a sister with polycystic ovarian syndrome (PCOS). The patient denies current or past use of alcohol, recreational drugs, or smoking. She reports that she has been on oral birth control pills since age 14 and uses condoms inconsistently. No history of STDs. She is sexually active with her current boyfriend, who was treated for chlamydia 2 years ago. She received and completed the HPV vaccination series starting at age 11. Her vital signs include: temperature

36.8°C (98.2°F), pulse 97/min, respiratory rate 16/min, blood pressure 120/75 mm Hg. Physical examination is unremarkable. Which of the following are the recommended guidelines for cervical cancer screening for this patient at this time?

Answer Choices:

A. Cytology (pap smear) every 3 years B. Cytology (pap smear) and HPV DNA co-testing every 5 years C. No cervical cancer screening is indicated at this time D. Cytology (pap smear) and HPV DNA co-testing every 3 years

Correct Answer: C. No cervical cancer screening is indicated at this time

Question 175

Clinical Scenario:

A 43-year-old man presents to the emergency department with nausea and vomiting. He says symptoms onset 4 hours ago and is progressively worsening. He denies any hematemesis. Past medical history is significant for a recent negative screening colonoscopy that was performed due to a family history of colon cancer. His vital signs are significant for a temperature of 39.5°C (103.1°F). Physical examination is unremarkable. A contrast CT of the abdomen reveals a colonic perforation. Laboratory findings are significant for an elevated WBC count with a predominant left shift, a decreased platelet count, increased PT and PTT, slightly decreased hemoglobin/hematocrit, and prolonged bleeding time. Which of the following is most closely related to this patient's prolonged bleeding time?

Answer Choices:

A. GpIIb/IIIa B. Vitamin K C. Fibrinogen D. Giant platelets

Correct Answer: C. Fibrinogen

Question 176

Clinical Scenario:

A 28-year-old man presents with a yellow coloration of his skin. He says he feels well and denies any recent history of nausea, fatigue or fever, or discoloration of his urine or stool. The patient reports episodes with similar symptoms in the past. Family history is significant for similar symptoms in his father. The patient is afebrile and vital signs are within normal limits. On physical examination, he is jaundiced. Scleral icterus is present. Laboratory findings are significant only for an unconjugated hyperbilirubinemia. Liver enzymes are normal, and there is no bilirubin present in the urine. Which of the following is the most appropriate treatment for this patient's most likely diagnosis?

Answer Choices:

A. Inhibitors of heme oxygenase B. Phenobarbital C. No therapy indicated D. Plasma exchange transfusion

Correct Answer: C. No therapy indicated

Question 177

Clinical Scenario:

During a study on chronic obstructive pulmonary disease (COPD), researchers discovered an agent that markedly inhibits the carbon dioxide-carrying capacity of the venous blood. Which of the following is the most likely mechanism underlying this agent's effects?

Answer Choices:

A. Decreased capillary permeability to carbon dioxide B. Decreased amount of dissolved plasma carbon dioxide C. Inhibition of erythrocyte carbonic anhydrase D. Increased solubility of carbon dioxide in plasma

Correct Answer: C. Inhibition of erythrocyte carbonic anhydrase

Question 178

Clinical Scenario:

A 4-year-old boy is brought to the physician for the evaluation of fatigue since he returned from visiting family in South Africa one week ago. The day after he returned, he had fever, chills, and diffuse joint pain for 3 days. His symptoms improved with acetaminophen. He was born at term and has been healthy. His immunizations are up-to-date. His temperature is 37.6°C (99.68°F), pulse is 100/min, and blood pressure is 100/60 mm Hg. Examination shows conjunctival pallor. The remainder of the examination shows no abnormalities. Laboratory studies show: Hemoglobin 10.8 g/dL Mean corpuscular volume 68 μm^3 Red cell distribution width 14% (N = 13%–15%) Hemoglobin A2 6% (N < 3.5%) A peripheral smear shows microcytic, hypochromic erythrocytes, some of which have a darkly stained center and peripheral rim, separated by a pale ring. Which of the following is the most appropriate next step in the management of this patient?"

Answer Choices:

A. Folic acid therapy B. Oral succimer C. Reassurance D. Iron supplementation

Correct Answer: C. Reassurance

Question 179

Clinical Scenario:

A 50-year-old man presents to his primary care provider complaining of double vision and trouble seeing out of his right eye. His vision started worsening about 2 months ago and has slowly gotten worse. It is now severely affecting his quality of life. Past medical history is significant for poorly controlled hypertension and hyperlipidemia. He takes amlodipine, atorvastatin, and a baby aspirin every day. He smokes 2–3 cigarettes a day and drinks a glass of wine with dinner every night. Today, his blood pressure is 145/85 mm Hg, heart rate is 90/min, respiratory rate is 14/min, and temperature is 37.0°C (98.6°F). On physical exam, he appears pleasant and talkative. His heart has a regular rate and rhythm and his lungs are clear to auscultation bilaterally. Examination of the eyes reveals a dilated left pupil that is positioned inferolateral with ptosis. An angiogram of

the head and neck is performed and he is referred to a neurologist. The angiogram reveals a 1 cm berry aneurysm at the junction of the posterior communicating artery and the posterior cerebral artery compressing the ocular nerve. Which of the following statements best describes the mechanism behind the oculomotor findings seen in this patient?

Answer Choices:

A. The parasympathetic nerve fibers of this patient's eye are activated. B. The unopposed superior oblique muscle rotates the eye downward. C. The unopposed inferior oblique muscle rotates the eye downward. D. The sympathetic nerve fibers of this patient's eye are inhibited.

Correct Answer: B. The unopposed superior oblique muscle rotates the eye downward.

Question 180

Clinical Scenario:

A 40-year-old man presents with acute abdominal pain. Past medical history is significant for hepatitis C, complicated by multiple recent visits with associated ascites. His temperature is 38.3°C (100.9°F), heart rate is 115/min, blood pressure is 88/48 mm Hg, and respiratory rate is 16/min. On physical examination, the patient is alert and in moderate discomfort. Cardiopulmonary examination is unremarkable. Abdominal examination reveals distant bowel sounds on auscultation. There is also mild diffuse abdominal tenderness to palpation with guarding present. The remainder of the physical examination is unremarkable. A paracentesis is performed. Laboratory results are significant for the following: Leukocyte count 11,630/μL (with 94% neutrophils) Platelets 24,000/μL Hematocrit 29% Ascitic fluid analysis: Cell count 658 PMNs/μL Total protein 1.2 g/dL Glucose 24 mg/dL Gram stain Gram-negative rods Culture Culture yields growth of E. coli Which of the following is the next, best step in the management of this patient?

Answer Choices:

A. Surgical consultation B. Intravenous cefotaxime C. Serum lipase level D. Abdominal radiography and contrast CT of the abdomen

Correct Answer: B. Intravenous cefotaxime

Question 181

Clinical Scenario:

Three days after undergoing open surgery to repair a bilateral inguinal hernia, a 66-year-old man has new, intermittent upper abdominal discomfort that worsens when he walks around. He also has new shortness of breath that resolves with rest. There were no complications during surgery or during the immediate postsurgical period. Ambulation was restarted on the first postoperative day. He has type 2 diabetes mellitus, hypercholesterolemia, and hypertension. He has smoked one pack of cigarettes daily for 25 years. Prior to admission, his medications included metformin, simvastatin, and lisinopril. His temperature is 37°C (98.6°F), pulse is 80/min, respirations are 16/min, and blood pressure is 129/80 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. The abdomen is soft and shows two healing surgical scars with moderate

serous discharge. Cardiopulmonary examination shows no abnormalities. An ECG at rest shows no abnormalities. Cardiac enzyme levels are within the reference range. An x-ray of the chest and abdominal ultrasonography show no abnormalities. Which of the following is the most appropriate next step in diagnosis?

Answer Choices:

A. Magnetic resonance imaging of the abdomen B. Culture swab from the surgical site C. Coronary angiography D. Cardiac pharmacological stress test

Correct Answer: D. Cardiac pharmacological stress test

Question 182

Clinical Scenario:

An 85-year-old man who recently immigrated to the US from Spain presents to your office complaining of hoarseness and dysphagia for the past 2 months. He says his symptoms have been getting progressively worse, and he expresses concerns about the difficulty swallowing as he cannot eat well and has even lost 9 kg (20 lb) since his last visit 3 months ago. He denies any shortness of breath, coughing of blood, and chest pain. His bowel and bladder habit are normal. Past medical history is unremarkable. He has a 60-pack-year history of smoking tobacco and drinks alcohol occasionally. Which of the following is the most likely cause of his symptoms?

Answer Choices:

A. Malignant proliferation of squamous cells B. Malignant proliferation of glandular tissue in the esophagus C. Reduced lower esophageal pressure D. Chronic autoimmune gastritis

Correct Answer: A. Malignant proliferation of squamous cells

Question 183

Clinical Scenario:

A 25-year-old man is in the middle of an ascent up a mountain, at an elevation of about 4,500 meters. This is the 4th day of his expedition. His friend notices that in the last few hours, he has been coughing frequently and appears to be short of breath. He has used his albuterol inhaler twice in the past 4 hours, but it does not seem to help. Within the past hour, he has coughed up some frothy, slightly pink sputum and is now complaining of nausea and headache. Other than his asthma, which has been well-controlled on a steroid inhaler, he is healthy. Which of the following is the most likely cause of this man's symptoms?

Answer Choices:

A. Pulmonary embolism B. Non-cardiogenic pulmonary edema C. An acute asthma exacerbation D. Pneumothorax

Correct Answer: B. Non-cardiogenic pulmonary edema

Question 184

Clinical Scenario:

A 55-year-old male bodybuilder presents to the emergency department with weakness of his right arm. The patient states he has experienced these symptoms for a few weeks; however, today his hand felt so weak he dropped his cup of tea. The patient has a past medical history of diabetes. He drinks 2-7 alcoholic drinks per day and has smoked 2 packs of cigarettes per day since he was 25. The patient admits to using anabolic steroids. He has lost 17 pounds since he last came to the emergency department 1 month ago. His temperature is 99.5°F (37.5°C), blood pressure is 177/108 mmHg, pulse is 90/min, respirations are 17/min, and oxygen saturation is 98% on room air. Physical exam reveals decreased sensation in the right arm and 2/5 strength in the right arm and 5/5 strength in the left arm. The patient states that he is experiencing a dull aching and burning pain in his right arm during the exam. Which of the following is the most likely diagnosis?

Answer Choices:

A. Apical lung tumor B. Cerebral infarction C. Scalenus anticus syndrome D. Subclavian steal syndrome

Correct Answer: A. Apical lung tumor

Question 185

Clinical Scenario:

A 5-year-old girl with an aortic stenosis correction comes to the office for a follow-up visit for acute lymphoblastic lymphoma. She initiated chemotherapy a week before through a peripherally inserted central line. She reports being 'tired all the time' and has been bruising easily. Her vital signs are within normal limits. Physical examination shows several tenders, non-blanching petechiae on the pads of the fingers and toes; several dark, non-tender petechiae on her palms and soles; and small, linear hemorrhages under her fingernails. Fundoscopic examination shows various small areas of hemorrhage on the retinae bilaterally. Cardiac examination is notable for a II/VI systolic ejection murmur that seems to have worsened in comparison to the last visit. Which of the following is the most likely cause?

Answer Choices:

A. Bleeding diathesis secondary to thrombocytopenia B. Dilated cardiomyopathy C. Hypertrophic cardiomyopathy D. Infective endocarditis

Correct Answer: D. Infective endocarditis

Question 186

Clinical Scenario:

A 4-year-old boy is brought to the physician because of a progressive headache and neck pain for 2 weeks. During this period, he has had multiple episodes of dizziness and tingling sensations in his arms and hands. A year ago, he underwent closed reduction of a dislocated shoulder that he suffered after a fall. He underwent

surgical removal of a sac-like protuberance on his lower back, soon after being born. His temperature is 36.7°C (98°F), pulse is 80/min, and blood pressure is 100/80 mm Hg. His neck is supple. Neurological examination shows sensorineural hearing loss bilaterally and normal gross motor function. Fundoscopy reveals bilateral optic disk swelling. An MRI of the brain is shown. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. Medulloblastoma B. Intraventricular hemorrhage C. Chiari II malformation D. Vestibular schwannoma

Correct Answer: C. Chiari II malformation

Question 187

Clinical Scenario:

Three hours after the onset of labor, a 39-year-old woman, gravida 2, para 1, at 40 weeks' gestation has sudden worsening of abdominal pain and vaginal bleeding. 18 months ago her first child was delivered by a lower segment transverse cesarean section because of cephalopelvic disproportion. Her temperature is 37.5°C (99.5°F), pulse is 120/min, respirations are 20/min, and blood pressure is 90/50 mm Hg. Examination shows abdominal tenderness and the absence of uterine contractions. The cervix is 100% effaced and 10 cm dilated; the vertex is at -3 station. An hour before, the vertex was at 0 station. Cardiotocography shows fetal bradycardia, late decelerations, and decreased amplitude of uterine contractions. Which of the following is the most specific feature of this patient's condition?

Answer Choices:

A. Loss of fetal station B. Hemodynamic instability C. Fetal distress D. Abdominal tenderness

Correct Answer: A. Loss of fetal station

Question 188

Clinical Scenario:

A 27-year-old woman presents to your office complaining of difficulty swallowing, and she describes that "there is something in the back of her throat". Furthermore, she also feels an "achy" chest pain that has been getting progressively worse over the last few weeks. She denies having any fever, shortness of breath, cough, abdominal pain, heartburn, nausea, or vomiting. The patient has a history of wrist fracture as a child, migraines, and a recent diagnosis of myasthenia gravis. Which of the following is the most likely diagnosis?

Answer Choices:

A. Benign tumor of the thymus B. Superior vena cava syndrome C. Anaplastic thyroid cancer D. Mediastinitis

Correct Answer: A. Benign tumor of the thymus

Question 189

Clinical Scenario:

A 16-year-old boy with a history of severe, persistent asthma presents to the emergency department with severe shortness of breath and cough. He states that he was outside playing basketball with his friends, forgot to take his inhaler, and began to have severe difficulty breathing. On exam, he is in clear respiratory distress with decreased air movement throughout all lung fields. He is immediately treated with beta-agonists which markedly improve his symptoms. Prior to treatment, which of the following was most likely observed in this patient?

Answer Choices:

A. Inspiratory stridor B. Increased breath sounds C. Friction rub D. Pulsus paradoxus

Correct Answer: D. Pulsus paradoxus

Question 190

Clinical Scenario:

A 29-year-old woman, gravida 1, para 0 at 11 weeks' gestation comes to the physician because of a 2-day history of left lower extremity pain and swelling. Her temperature is 37.9°C (100.2°F). Physical examination shows a tender, palpable cord on the lateral aspect of the left lower leg. The overlying skin is erythematous and indurated. Duplex ultrasound shows vascular wall thickening and subcutaneous edema. Which of the following is the most likely diagnosis?

Answer Choices:

A. Deep vein thrombosis B. Erythema nodosum C. Varicose vein D. Superficial thrombophlebitis

Correct Answer: D. Superficial thrombophlebitis

Question 191

Clinical Scenario:

A 55-year-old man presents to his primary care physician for a new patient appointment. The patient states that he feels well and has no concerns at this time. The patient has a past medical history of hypertension, an elevated fasting blood glucose, and is not currently taking any medications. His blood pressure is 177/118 mmHg, pulse is 90/min, respirations are 16/min, and oxygen saturation is 97% on room air. Physical exam is notable for an obese man with atrophy of his limbs and striae on his abdomen. Laboratory values are notable for a blood glucose of 175 mg/dL. Which of the following is the best initial step in management?

Answer Choices:

A. Dexamethasone suppression test B. Hydrochlorothiazide C. Metformin D. MRI of the head

Correct Answer: A. Dexamethasone suppression test

Question 192

Clinical Scenario:

An experimental new drug (SD27C) is being studied. This novel drug delivers insulin via the intranasal route. Consent is obtained from participants who are diabetic and are taking insulin as their current treatment regimen to participate in a clinical trial. 500 patients consent and are divided into 2 groups, and a double-blind clinical trial was conducted. One group received the new formulation (SD27C), while the second group received regular insulin via subcutaneous injection. The results showed that the treatment outcomes in both groups are the same. SD27C is currently under investigation in which phase of the clinical trial?

Answer Choices:

A. Phase IV B. Phase III C. Post-market surveillance D. Phase I

Correct Answer: B. Phase III

Question 193

Clinical Scenario:

An 85-year-old woman presents to her physician with complaints of significant weakness and weight loss. She recently has been diagnosed with stage IV breast cancer for which she currently is receiving treatment. She mentions that, despite taking a diet rich in protein and calories, she continues to lose weight. On physical examination, her vital signs are stable, but muscle wasting is clearly evident in her upper limbs, lower limbs, and face. The physician explains to her that her advanced cancer is the most important cause for the weight loss and muscle wasting. This cachexia is mediated by the proteolysis-inducing factor released from cancer cells. Which of the following effects is produced by this factor?

Answer Choices:

A. Activation of hormone-sensitive lipase in adipose tissue B. Increased release of tumor necrosis factor (TNF) from macrophages C. Activation of NF- κ B D. Suppression of the appetite center in the hypothalamus

Correct Answer: C. Activation of NF- κ B

Question 194

Clinical Scenario:

A 73-year-old man comes to the physician because of worsening bilateral lower extremity pain for the past 8 months. The pain begins after walking one to two blocks and radiates bilaterally down the buttocks with cramping and tingling. He reports that the pain is worse while walking downhill and is relieved by sitting and leaning forward. He has hypertension, hyperlipidemia, and type 2 diabetes mellitus. He had a myocardial infarction at the age of 55 years and an abdominal aortic aneurysm repair at the age of 60 years. He has smoked one pack of cigarettes daily for the past 30 years. He does not drink alcohol or use illicit drugs. His current

medications include sitagliptin, metformin, atorvastatin, metoprolol succinate, amlodipine, and hydrochlorothiazide. His temperature is 37.5°C (99.5°F), pulse is 82/min, respirations are 17/min, and blood pressure is 150/87 mm Hg. Examination shows full muscle strength. Sensation is reduced bilaterally in the feet and toes. Straight leg raise is negative. X-ray of the spine shows degenerative changes. Which of the following is the most appropriate next step in diagnosis?

Answer Choices:

A. Measurement of HLA-B27 antigen B. Measurement of the ankle brachial index C. MRI scan of the spine D. Polysomnography

Correct Answer: C. MRI scan of the spine

Question 195

Clinical Scenario:

A 47-year-old female with a history of mild asthma, type II diabetes, hypertension, and hyperlipidemia presents to clinic complaining of swelling in her lips (Image A). She has had no changes to her medications within the past two years. Vital signs are stable. Physical exam is notable for significant erythema around and swelling of the lips. The remainder of her exam is unremarkable. What is the mechanism of action of the drug that has caused her current symptoms?

Answer Choices:

A. Inhibition of angiotensin-converting enzyme B. Inhibition of HMG-CoA reductase C. Stimulation of the Beta 2 receptor D. Inhibition of the Na/K/Cl triple transporter of the thick ascending limb

Correct Answer: A. Inhibition of angiotensin-converting enzyme

Question 196

Clinical Scenario:

A group of scientists is verifying previous research on DNA replication. The diagram below illustrates the theoretical DNA replication process in bacteria such as E. coli. Which of the following enzymes would need to be decreased to prevent the removal of RNA primers formed in the lagging strand?

Answer Choices:

A. Helicase B. DNA polymerase I 3' to 5' exonuclease activity C. DNA polymerase I 5' to 3' exonuclease activity D. DNA ligase

Correct Answer: C. DNA polymerase I 5' to 3' exonuclease activity

Question 197

Clinical Scenario:

A 38-year-old man presents with concerns after finding out that his father was recently diagnosed with colon cancer. Family history is only significant for his paternal grandfather who also had colon cancer. A screening colonoscopy is performed, and a polyp is found in the ascending (proximal) colon, which on biopsy shows adenocarcinoma. A mutation in a gene that is responsible for which of the following cellular functions is the most likely etiology of this patient's cancer?

Answer Choices:

A. Inhibitor of apoptosis B. Inhibits progression from G1 to S phase C. DNA mismatch repair D. RAS cycle transduction inhibitor

Correct Answer: C. DNA mismatch repair

Question 198

Clinical Scenario:

A 33-year-old woman presents to her local clinic in rural eastern India complaining of neck pain and fever. She reports a 4 day history of severe neck pain, neck stiffness, mild diarrhea, and fever. She has not taken her temperature. She works as a laborer and frequently carries heavy weights on her back. She is prescribed a medication and told to come back if her symptoms do not improve. Her symptoms resolve after a couple days. Six months later, she gives birth to a newborn male at 34 weeks gestation. His temperature is 97.8°F (36.6°C), blood pressure is 90/55 mmHg, pulse is 110/min, and respirations are 24/min. On examination, the baby is irritable with a weak cry. Ashen gray cyanosis is noted diffusely. What is the mechanism of action of the drug responsible for this child's presentation?

Answer Choices:

A. DNA-dependent RNA polymerase inhibitor B. DNA gyrase inhibitor C. 30S ribosomal subunit inhibitor D. 50S ribosomal subunit inhibitor

Correct Answer: D. 50S ribosomal subunit inhibitor

Question 199

Clinical Scenario:

An 11-month-old boy presents with the recent loss of appetite and inability to gain weight. His diet consists mainly of cow's milk and fruits. Family history is unremarkable. Physical examination shows conjunctival pallor. Laboratory findings are significant for the following: Hemoglobin 9.1 g/dL Mean corpuscular volume 75 μm^3 Mean corpuscular hemoglobin 20 pg/cell Red cell distribution width 18% The patient is presumptively diagnosed with iron deficiency anemia (IDA) and ferrous sulfate syrup is prescribed. Which of the following laboratory values would most likely change 1st in response to this treatment?

Answer Choices:

A. $\uparrow$ reticulocyte count B. Anisocytosis C. $\downarrow$ mean corpuscular hemoglobin D. $\downarrow$ Mentzer index

Correct Answer: A. $\uparrow$ reticulocyte count

Question 200

Clinical Scenario:

A 36-year-old man presents to his physician with an acute burning retrosternal sensation with radiation to his jaw. This sensation began 20 minutes ago when the patient was exercising at the gym. It does not change with position or with a cough. The patient's vital signs include: blood pressure is 140/90 mm Hg, heart rate is 84/min, respiratory rate is 14/min, and temperature is 36.6°C (97.9°F). Physical examination is only remarkable for paleness and perspiration. The patient is given sublingual nitroglycerin, the blood is drawn for an express troponin test, and an ECG is going to be performed. At the moment of performing ECG, the patient's symptoms are gone. ECG shows increased R amplitude in I, II V3-V6, and ST depression measuring for 0.5 mm in the same leads. The express test for troponin is negative. Which of the following tests would be reasonable to perform next to confirm a diagnosis in this patient?

Answer Choices:

A. Blood test for CPK-MB B. Chest radiography C. CT angiography D. Exercise stress testing

Correct Answer: D. Exercise stress testing

Question 201

Clinical Scenario:

A 54-year-old woman comes to the emergency department because of two episodes of bright red blood per rectum within the past day. She has a history of migraine, which is treated prophylactically with verapamil. She appears well and is hemodynamically stable. Cardiac exam reveals a regular heart rate without any murmurs or gallops. Lungs are clear to auscultation. Her abdomen is mildly tender without rebound or guarding. Digital rectal examination shows fresh blood on the glove. Laboratory studies show: Hemoglobin 10.4 g/dL Leukocyte count 5,000/mm³ Platelet count 175,000/mm³ Partial thromboplastin time 35 seconds Serum Na⁺ 140 mEq/L K⁺ 3.7 mEq/L Cl⁻ 101 mEq/L HCO₃⁻ 25 mEq/L Mg²⁺ 1.8 mEq/L A routine ECG shows a heart rate of 75/min, a normal axis, PR interval of 280 ms, QRS interval of 80 ms with a QRS complex following each p wave, and no evidence of ischemic changes. Which of the following is the most appropriate next step in management with respect to this patient's cardiovascular workup?"

Answer Choices:

A. Observation B. Synchronized cardioversion C. Metoprolol therapy D. Pacemaker placement

Correct Answer: A. Observation

Question 202

Clinical Scenario:

A 64-year-old man presents to the office for an annual physical examination. He has no complaints at this visit. His chart states that he has a history of hypertension, chronic obstructive pulmonary disease (emphysema), Raynaud's disease, and glaucoma. He is a 30 pack-year smoker. His medications included lisinopril, tiotropium, albuterol, nifedipine, and latanoprost. The blood pressure is 139/96 mm Hg, the pulse is 86/min, the respiration rate is 16/min, and the temperature is 37.2°C (99.1°F). On physical examination, his pupils are equal, round, and reactive to light. The cardiac auscultation reveals an S4 gallop without murmur, and the lungs are clear to auscultation bilaterally. However, the inspection of the chest wall shows an enlarged anterior to posterior diameter. Which of the following is the most appropriate screening test for this patient?

Answer Choices:

A. Low-dose CT B. Magnetic resonance imaging C. Bronchoalveolar lavage with cytology D. Pulmonary function tests

Correct Answer: A. Low-dose CT

Question 203

Clinical Scenario:

A 24-year-old woman presents to the clinic with chronic abdominal discomfort and cramping. She seeks medical attention now as she is concerned about the diarrhea that she has developed that is occasionally mixed with tiny streaks of blood. Her medical history is significant for lactose intolerance and asthma. She has a family history of wheat allergy and reports that she has tried to make herself vomit on several occasions to lose weight. After counseling the patient about the dangers of bulimia, physical examination reveals the rectum is red, inflamed, tender, and a perirectal abscess is seen draining purulent material. Colonoscopy demonstrates scattered mucosal lesions involving the colon and terminal ileum. A complete blood count is given below:

Hb%: 10 gm/dL Total count (WBC): 12,500/mm³ Differential count:

Neutrophils: 50% Lymphocytes: 40% Monocytes: 5% ESR: 22 mm/hr What is the most likely diagnosis?

Answer Choices:

A. Irritable bowel syndrome B. Celiac disease C. Ulcerative colitis D. Crohn's disease

Correct Answer: D. Crohn's disease

Question 204

Clinical Scenario:

An 86-year-old woman is brought to the emergency department by her niece because the patient felt like she was spinning and about to topple over. This occurred around 4 hours ago, and although symptoms have improved, she still feels like she is being pulled to the right side. The vital signs include: blood pressure 116/75 mm Hg, pulse 90/min, and SpO₂ 99% on room air. Physical examination reveals right-sided limb ataxia along with hypoalgesia and decreased temperature sensation on the right side of the face and left side of the body. An urgent non-contrast CT scan of the head shows no evidence of hemorrhage. What other finding is most likely to be present in this patient?

Answer Choices:

A. Hemiparesis B. Hemianopia C. Intact cough reflex D. Absent gag reflex

Correct Answer: D. Absent gag reflex

Question 205**Clinical Scenario:**

An 8-year-old boy is brought to the pediatrician by his parents due to recurrent episodes of wheezing for the last 2 years. He uses a salbutamol inhaler for relief from wheezing, but his symptoms have recently worsened. He often coughs during the night, which awakens him from sleep almost every other day. He is not able to play football because he starts coughing after 10–15 minutes of play. His current physical examination is completely normal and auscultation of his chest does not reveal any abnormal breath sounds. His peak expiratory flow rate (PEFR) is 75% of expected for his age, gender, and height. After a complete diagnostic evaluation, the pediatrician prescribes a low-dose inhaled fluticasone daily for at least 3 months. He also mentions that the boy may require continuing inhaled corticosteroid (ICS) therapy for a few years if symptoms recur after discontinuation of ICS. However, the parents are concerned about the side effects of corticosteroids. Which of the following corticosteroid-related adverse effects is most likely?

Answer Choices:

A. Suppression of hypothalamus-pituitary-adrenal (HPA) axis B. Steroid psychosis C. Hoarseness of voice D. Short stature

Correct Answer: C. Hoarseness of voice

Question 206**Clinical Scenario:**

A 38-year-old woman undergoes hemithyroidectomy for treatment of localized, well-differentiated papillary thyroid carcinoma. The lesion is removed with clear margins. However, during the surgery, a structure lying directly adjacent to the superior thyroid artery at the upper pole of the thyroid lobe is damaged. This patient is most likely to experience which of the following symptoms?

Answer Choices:

A. Voice pitch limitation B. Ineffective cough C. Weakness of shoulder shrug D. Shortness of breath

Correct Answer: A. Voice pitch limitation

Review Summary

Category: Diagnosis **Questions Reviewed:** _____ / 206

Overall Assessment: - Correctly categorized: _ **questions** - **Miscategorized:** *questions* - **Ambiguous:** __
questions

General Comments:

Reviewer Name: _____

Date: _____

Credentials: _____